A Complete History of the Earth's Sky

What could be seen from the ground at nineteen moments in Earth's history: the color of the sky, straight up and at the horizon, at the equator and the poles, at noon and at dusk, reconstructed with a spectral radiative-transfer model; and the Sun, Moon, planets, stars and everything else above it, from the molten newborn Moon to a sky full of satellites.

James Guillochon · guillochon@gmail.com · October 2026

Interactive experience at earthsky.astrocrash.net

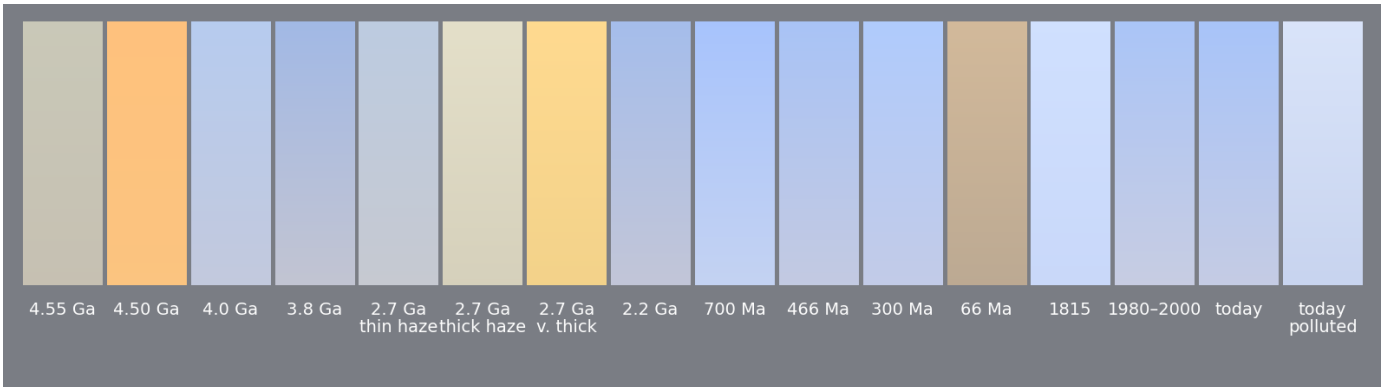


Figure 1. Mid-latitude noon sky at the epochs, zenith (top) to horizon (bottom), oldest at left. Brightness shown relative to today's clean sky.

Ten things the view did

1. Under the 30-bar CO_2 of the early Hadean the sky was a shadowless peach-white glow with no visible Sun, and the molten newborn Moon, 4° across and 8 Earth radii away, glowed orange through it, rising and setting several times in a 6-hour day.
2. During the hazy Neoarchean the whole dome was cream to apricot, with almost no difference between zenith and horizon.
3. Before the Great Oxidation every twilight ended in a cream zenith; the first ozone turned it pale blue, and the deep blue dusk needs a near-modern column.
4. Snowball Earth had the bluest mid-latitude sky in Earth's history; the K–Pg impact winter had no sunsets at all.
5. The eclipsed Moon is copper today, orange-rimmed before there was ozone, dark-cored when the Moon was close, and gone under the K–Pg soot, the thick Archean haze or a Tambora veil.
6. Before the Great Oxidation the nights were darker and violet: no oxygen airglow, nitrogen aurorae reaching the mid-latitudes, and meteors with no glowing trains.
7. The young Sun was smaller, redder and dimmer, and so heavily spotted that a spot could be seen on it every day.
8. The Moon's face was not always today's: at 3.8 Ga the maria were still flooding, and at 4.0 Ga the near side was bright highland from limb to limb.
9. Saturn had no rings before roughly 100 to 200 million years ago, and θ Columbae, now a faint fifth-magnitude star, shone as bright as Venus 4.7 million years ago.
10. In 2100 thousands of orbital datacenters may outshine nearly every star at dusk, in a bright band along the terminator.

1 Introduction

What did the sky look like from the ground in the Archean, under the young Sun, with the Moon nearer? We reconstruct the whole view from Earth's surface at nineteen moments in its history, from a proto-Earth 17 million years after the first solids to the year 2100: the color of the sky by day and at dusk, computed with a spectral radiative-transfer model of each epoch's air, and on that sky everything else that could be seen from the ground: the Sun and its spots, the Moon's size, face and eclipses, the planets and their moons, the stars and the Galaxy as they then lay, meteors, comets, aurorae, airglow, halos and clouds, and in the latest epochs the light people added. It began as a study of the sky's color, which remains its quantitative core, and grew with the interactive site built on the model, which draws each of these and is the source of every number quoted here outside the color tables.

The nineteen epochs (Table 1) were chosen for what changed: the air (a 30-bar CO_2 Hadean, an Archean with no oxygen and organic haze, the first ozone after the Great Oxidation, a frozen Snowball, the oxygen-rich Carboniferous, the impact winter after Chicxulub, a volcanic year, the ozone hole and today's cities), the sky above it (a Moon forming and receding, a young and active Sun, a Solar System before the giant planets' instability, nearby supernovae, stars that outshone Sirius, a bombardment and a possible ring), and the view people have made (city light and a sky full of satellites). Each is a reconstruction, with the uncertainties the sections state: some quantities are measured, some modeled, some estimated to show a trend, and some, like where the planets stood on a given day four billion years ago, unknowable and drawn at random.

This is a living document. Much of what it draws rests on questions that are still argued: when the giant planets went unstable and whether a fifth giant was lost; how far away the Moon was in its first hundred million years; how thick the Archean haze was and how much ozone the first oxygen made; how old Saturn's rings are; whether the Earth had a ring in the Ordovician; how bright the young Sun's corona and wind were. Where the literature disagrees, we say which reading we take and why, and as the consensus moves the site and this report will move with it: epochs will be redrawn, numbers revised and new findings added. Each version carries its date, and the code and data that produce it are public, so any figure here can be traced to the model and the sources behind it.

The site computes what it shows as realistically as it can while still loading in seconds and running smoothly in a browser, and where the two conflict it compromises. Some compromises stand in for physics too costly to run. Jupiter's and Saturn's clouds are a tracer stirred by a flow made of noise, shaped by the measured winds and rolled into eddies and billows, a proxy for the planets' true three-dimensional hydrodynamics rather than a simulation of it; the animation repeats over a 2000-day cycle. The sky's multiple scattering is a two-stream approximation rather than a full Monte Carlo transfer, computed ahead of time on a grid of Sun heights and interpolated between them. The clouds are scenery, procedural volumes lit by the model's sky rather than the output of a climate model, and the Milky Way is a model of its structure rather than a photograph. The photosphere's granulation and the Moon's craters' shadows are drawn from procedural cells and a limited march toward the Sun, and the close-up detail of Jupiter, Saturn, the Sun and the spacecraft maps is built only when the view zooms in on them. Other compromises are made for the eye: the Sun and Moon are drawn at four times their size, the sunset's colors are spread further apart than they are, and the Sun's limb darkening is softened. Each section says where its subject is measured, modeled or approximated.

In Section 2 we describe how the view is reconstructed. Section 3 gives the sky's colors by day and at dusk, and Section 4 through Section 9 take in turn the Sun, the Moon, the planets, the stars and the Galaxy, the night and what crosses it, and the sky people made. Section 10 goes through the epochs one by one, and Section 11 says what the reconstruction leaves out and which of its numbers are estimates.

Table 1: The nineteen epochs. The Sun's light is relative to today's and its temperature from the solar track of Baraffe et al. (2015); the day and the Moon's width from the Moon's distance (Table 5); the last column lists what the sections that follow find in each sky besides its color.

Age	Epoch	Sun's light	Sun (K)	Day	Moon's width	What else is in the sky
4.55 Ga	Proto-Earth	0.47	4,614	12.0 h	none	No Moon; Theia passing; the giant planets before the instability, with a fifth giant; a saturated young Sun
4.50 Ga	Early Hadean	0.68	5,532	5.9 h	7.52×	The molten Moon at 8 R⊕; 10,000× today's meteors; no stars through 30 bar of CO_2 and the debris's glow
4.0 Ga	Late Hadean	0.72	5,600	12.6 h	1.51×	Bombardment; nitrogen aurorae to 37°; the Moon before the maria
3.8 Ga	Early Archean	0.75	5,620	12.8 h	1.49×	The maria flooding, lava fountains on the Moon; Mars's lost moon Harmonia
2.7 Ga	Neoarchean (thin haze)	0.80	5,660	15.6 h	1.26×	Thin organic haze; an 18-month year
2.7 Ga	Neoarchean (thick haze)	0.80	5,660	15.6 h	1.26×	The Pale Orange Dot seen from below
2.7 Ga	Neoarchean (very thick haze)	0.80	5,660	15.6 h	1.26×	The thickest plausible haze
2.2 Ga	Paleoproterozoic	0.85	5,700	17.4 h	1.18×	The first ozone; oxygen airglow at 30%
700 Ma	Snowball Earth	0.94	5,750	21.9 h	1.04×	Ice halos at every latitude; the clearest air
466 Ma	Ordovician meteor storm	0.96	5,760	22.3 h	1.03×	100× today's fireballs and perhaps a ring
300 Ma	Carboniferous	0.97	5,765	22.6 h	1.03×	Saturn without rings, and its moon Chrysalis
66 Ma	K-Pg impact winter	0.99	5,772	23.8 h	1.00×	Soot from Chicxulub; Saturn's young rings
1.78 Ma	ζ Ophiuchi supernova	1.00	5,772	24.0 h	today's	A supernova at −11.6 in Scorpius; stars moved back
342 ka	Geminga supernova	1.00	5,772	24.0 h	today's	A supernova at −11 in Orion; Aldebaran at −1.5
1815	Volcanic year (Tambora 1815 / Toba-lite)	1.00	5,772	24.0 h	today's	Tambora's veil; oil lamps; no Red Spot yet
1980–2000	Ozone-hole spring	1.00	5,772	24.0 h	today's	Polar-spring ozone at 130 DU; sodium-lit cities
today	Modern, clean air	1.00	5,772	24.0 h	today's	A mid-sized city's glow; airliners; the real comets and showers
today	Modern, polluted megacity	1.00	5,772	24.0 h	today's	A polluted megacity
2100	Year 2100	1.00	5,772	24.0 h	today's	Megaconstellations and orbital datacenters

1.1 Has this been done before?

Not as such. A search of the Valency corpus (arXiv, PubMed, EarthArXiv and others) finds the two halves of the sky-color problem solved separately but never joined. On one side, atmospheric-optics work computes modern sky color rigorously: Lynch & Mazuk (2005) derive CIE chromaticities of the sky, Sun and distant objects from MODTRAN spectra; Peyvandi et al. (2016) simulate 5.6 million daylight spectra with SBDART to map how aerosol and ozone shift sky color; Lee & Mollner (2017) show with Monte Carlo simulation that vivid solar-horizon twilight colors need aerosol while the purest antisolar colors occur in a purely molecular atmosphere. On the other side, paleoatmosphere work constrains composition: Catling & Zahnle (2020) for the Archean (O_2 below 10^{-6} of present, CO_2 10–2,500× modern, CH_4 10^2–10^4× modern), and Arney et al. (2016), Mak et al. (2023) and Goldblatt et al. (2024) for the organic-haze episodes. Arney's "Pale Orange Dot" computes what a hazy Archean Earth looks like *from space*, as an exoplanet spectrum, and Sanromá & Pallé (2012) Earth's light curves over the last 500 million years; that is as close as the literature comes. Nobody has computed the view from the ground, or the horizon-versus-zenith, latitude, and sunset structure.

The same holds for the rest of the view. Its pieces are separate results: the Moon's recession and the day's lengthening (Farhat et al. 2022), the young Sun's brightness and activity, the stars' past motions, the giant planets' early instability. Earth's transmission spectrum has been computed through geological time as a transiting exoplanet would show it (Kaltenegger, Lin & Madden 2020), and today's lunar eclipse has been modeled with refraction (García Muñoz et al. 2012), but no reconstruction of how the eclipsed Moon itself looked in past epochs turns up, nor of how these pieces together looked from the ground on a given evening. We assemble them here.

2 How the view is reconstructed

2.1 The sky's colors

For each epoch the atmosphere is built from components with their own vertical profile and spectral optical depth: Rayleigh-scattering gas (N_2, O_2, CO_2, CH_4, H_2O in their epoch-appropriate partial pressures), ozone (the Chappuis cross section of Serdyuchenko et al. 2014 at 223 K, on a layer whose peak falls from 26 km at the equator to 18 km at the pole), tropospheric aerosol, a stratospheric sulfate layer, a tholin-like fractal organic haze (strongly blue-absorbing, forward-scattering), and impact soot (Table 2). Sky radiance is integrated numerically along each line of sight through a spherical-shell atmosphere, so grazing sunlight at twilight is handled properly, with multiple scattering and surface reflection added from a delta-Eddington two-stream solution. Spectra from 380–780 nm are converted to CIE XYZ with the analytic color-matching functions of Wyman, Sloan & Shirley (2013) and shown as sRGB. The model reproduces measured modern values (clear zenith 10,000–20,000 K, horizon 6,000–8,500 K, setting Sun ∼1,700 K, pink antisolar Belt of Venus) before being pointed backward.

We treat the young Sun as a blackbody whose effective temperature and luminosity follow the solar track of Baraffe et al. (2015): 5,530 K and 68% of today's output at 4.5 Ga, rising through 5,620 K (75%) at 3.8 Ga, 5,660 K (80%) at 2.7 Ga and 5,700 K (85%) at 2.2 Ga to 5,772 K today, and 4,614 K (47%) for the 17-million-year-old Sun of 4.55 Ga. The color effect is small: the 5,530 K Sun of 4.5 Ga is only about 240 K cooler in color temperature than the modern one, a difference that is barely perceptible side by side and is swamped by atmospheric effects ten to a hundred times larger. The brightness effect is not small, and is why the early-epoch skies are rendered dimmer. We ignore Fraunhofer lines and the young Sun's ultraviolet excess, which matter for haze photochemistry but not for visible color. The model is provided as the Python source *skymodel.py*; *run_epochs.py* produces the sky colors and *limb_grid.py* the limb and disk colors.

2.2 The rest of the view

The sky's colors come from the radiative-transfer model described above; the rest of the view is drawn on them by the interactive site, in two views: a dome, the whole sky as a fisheye, and a first-person view in WebGL, with a walkable ground and clouds. Both draw on one photometric scale, calibrated in absolute units: the clear zenith with the Sun 45° up is 2.7 kcd/m^2, which puts the full-Moon sky at the measured 18 mag/arcsec^2. Stars, planets, meteors and satellites are placed by their magnitudes on that scale and dimmed by the epoch's air, 0.15 magnitude per airmass from the gas plus 1.086 times its aerosol or haze optical depth; each shows only when it is brighter than the naked-eye limit for the sky around it. The Moon, moonlight, supernovae, fireballs and city glow light the sky through the same model. The clock runs over one solar day of the epoch, and the Sun follows the seasons for the date with today's 23.4° tilt in every epoch, since the Moon has held the obliquity between about 22° and 24.5°. The Sun and Moon are drawn at four times their angular size so that their disks show on a screen, and zooming in acts as a telescope, down to a field of 14 arcseconds.

Where each body was is carried back as far as it can be known. The Moon's distance follows the tidal record, the Sun's brightness and size its evolutionary track, the stars their measured motions back to the Pleistocene and orbits in a model Galaxy to 66 Ma; beyond that the star field and the planets' positions are drawn at random, with the right statistics, since no calculation can recover them. The colors are shown as a daylight-balanced camera would record them, without chromatic adaptation, and the night as a dark-adapted eye sees it. The scenery and the clouds are drawn, not modeled; the sky is the

Table 2: Atmospheres used for each epoch. Gas amounts are partial pressures in bar; AOD is tropospheric aerosol optical depth at 550 nm; haze, sulfate and soot are column optical depths at 550 nm; T_{eff} and L/L$\odot$ are the assumed solar effective temperature and luminosity relative to today. The supernova epochs and 2100 have today's clean air.

Epoch	Gas (bar)	O_3 (DU)	AOD	Haze	Sulfate	Soot	T_{eff} (K)	L/L$\odot$
4.55 Ga proto-Earth	CO_2 3, N_2 1, H_2O 0.05 (+dust 0.05)	0	0.15	–	–	–	4614	0.47
4.5 Ga Hadean	CO_2 30, N_2 1, H_2O 0.3	0	0.30	–	–	–	5532	0.68
4.0 Ga Hadean	CO_2 0.5, N_2 1, H_2O 0.02	0	0.15	–	–	–	5600	0.72
3.8 Ga Archean	N_2 0.8, CO_2 0.1, CH_4 0.001	0	0.08	–	–	–	5620	0.75
2.7 Ga thin haze	N_2 0.8, CO_2 0.05, CH_4 0.003	0	0.08	0.15	–	–	5660	0.8
2.7 Ga thick haze	N_2 0.8, CO_2 0.05, CH_4 0.005	0	0.08	0.6	–	–	5660	0.8
2.7 Ga very thick haze	N_2 0.8, CO_2 0.05, CH_4 0.01	0	0.08	1.5	–	–	5660	0.8
2.2 Ga post-GOE	N_2 0.8, O_2 0.002, CO_2 0.02	66	0.10	–	–	–	5700	0.85
700 Ma Snowball	N_2 0.78, O_2 0.02, CO_2 0.01 (+dust 0.05)	169	0.02	–	–	–	5750	0.94
466 Ma Ordovician	N_2 0.78, O_2 0.17, Ar 0.01, CO_2 0.0025 (+dust 0.01)	291	0.10	–	–	–	5760	0.962
300 Ma Carboniferous	N_2 0.78, O_2 0.33, Ar 0.01	323	0.15	–	–	–	5765	0.975
66 Ma impact winter	modern air	300	0.15	–	0.5	1.5	5772	0.995
1815 volcanic year	modern air	300	0.10	–	0.4	–	5772	1
Ozone-hole spring	modern air	130	0.10	–	–	–	5772	1
Modern, clean	N_2 0.78, O_2 0.21, Ar 0.01	300	0.10	–	–	–	5772	1
Modern, polluted	modern air	300	0.60	–	–	–	5772	1

model.

2.3 The ground

The ground is scenery, like the clouds: a walkable landscape chosen for each era rather than computed, lit by the model's sky and Sun. The proto-Earth has magma pools; the 4.5 Ga Earth a hot ocean, about 230 °C while 100 to 200 bars of CO_2 remained (Zahnle et al. 2010), steaming round volcanic islands; the Archean, water and volcanic shields; Snowball Earth, ice plateaus with blue banded cliffs and frozen ponds; the Ordovician, shallow seas with no land plants beyond mosses and crusts; the Carboniferous, forest and swamps; the impact winter, dead woods; the Pleistocene, 1815 and today, woods and snow-capped peaks; and the modern epochs, a town with lit windows. Water reflects the sky and glitters in the Sun and Moon; lava glows; distant scenery fades into the sky's color. None of this is part of the model: the sky above it is.

3 The sky by day and at dusk

3.1 Horizon versus zenith

In a clean molecular atmosphere the zenith is blue and the horizon white because the horizon path is long enough that blue is scattered back out before reaching the eye and multiple scattering fills in the other colors. The size of that gradient is the clearest single diagnostic of atmospheric state. It is largest in the cleanest atmospheres (Snowball Earth, today's clean air: 6,500–12,000 K of color-temperature difference), shrinks as Rayleigh depth rises (late Hadean), collapses to nothing when a broadband absorber or thick scatterer sits overhead (hazy Archean, impact winter, 30-bar Hadean), and actually *reverses* under a stratospheric sulfate veil or heavy pollution, where the zenith becomes whiter than the horizon.

3.2 Equator, mid-latitudes, poles

Latitude enters through solar elevation and surface albedo. A high Sun lights the sky from above through a short path, giving a brighter but paler zenith (equatorial noon ∼9,000 K); a low Sun sends light through more air, so the polar summer zenith is a deeper, darker blue (∼19,000 K today, 18,500 K over Snowball ice) while the polar horizon is warmer and yellower. The same geometry makes the poles the place where any overhead absorber is most visible: the hazy Archean sky is cream at the equator but distinctly orange-yellow at the poles, and the impact-winter Sun that is still a pale disk at the equator is gone at 75° latitude. Snow and ice brighten and blue-shift the whole dome by reflecting light back into the atmosphere, which is why polar and Snowball horizons stay blue where a mid-latitude land horizon goes white.

3.3 Particulates

Three kinds of particle appear in the record, and each leaves a different sky. Non-absorbing sulfate high in the stratosphere (volcanic years) whitens the day sky and produces long, luminous pink and lavender twilights, because the layer is still sunlit after the ground is dark. Absorbing organic haze (Neoarchean) or soot (impact winter) removes blue outright, turning the dome tan or amber with almost no directional structure, and dims the surface by factors of 3–20. Boundary-layer pollution or dust brightens and grays the zenith, and buries the setting Sun in murk before it reaches the horizon. Tropospheric aerosol is also what makes sunset color vivid: the model confirms the result of Lee & Mollner (2017) that a purely molecular sky yields a pale, yellow-orange horizon while any turbidity deepens it to red.

3.4 Sunsets and sunrises through time

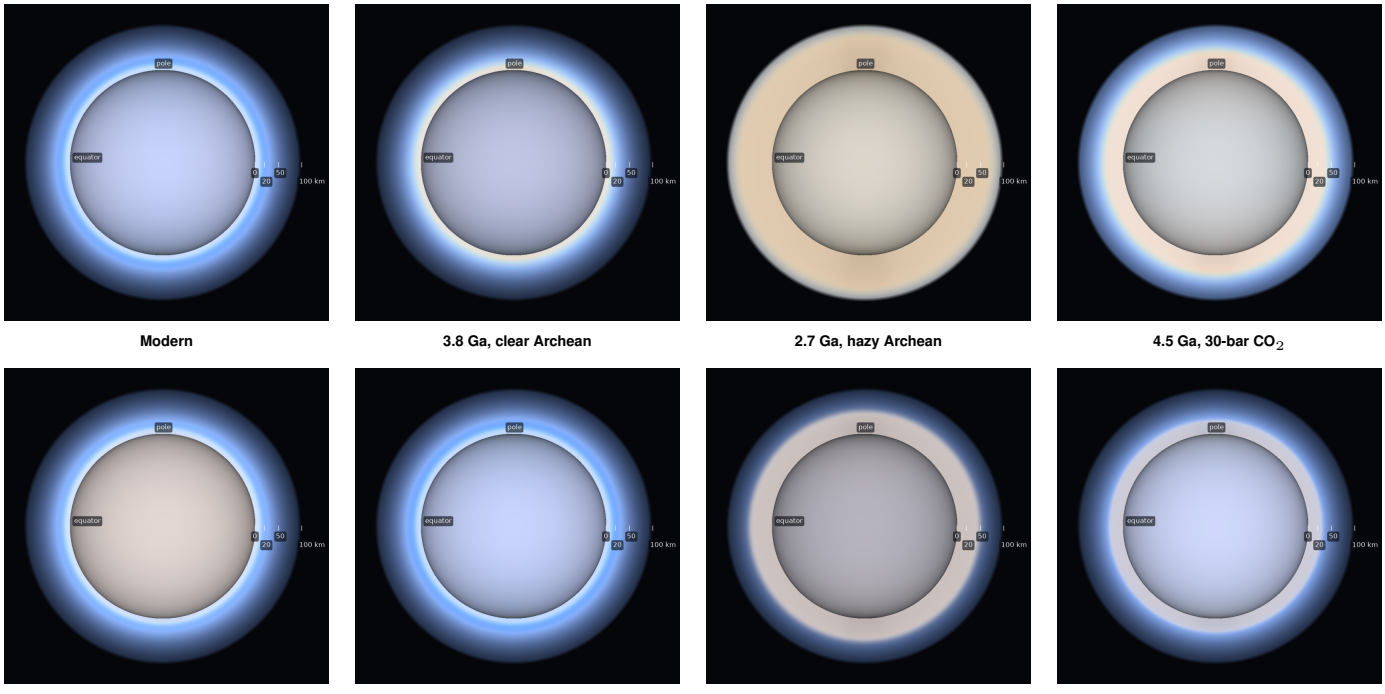
The biggest surprise is ozone. Today the sky above a setting Sun grades from orange through pale yellow to a zenith that stays blue well into twilight: about 12,500 K with the Sun 4° down. That persistent blue is the Chappuis band, ozone absorbing yellow and orange along the grazing path. The same modern sky with the ozone removed drops that zenith to cream, about 6,000 K. Before the Great Oxidation Event there was no ozone, so every sunset for the first two billion years ended with the zenith fading through cream and beige to orange. At ∼2.2 Ga a 66 DU column, the Cooke et al. (2022) global mean for 1% of present oxygen, turns that cream zenith a pale blue-gray near 7,100 K. The deep twilight blue arrives later, once the column is near modern: the snowball sky, at their 10% oxygen column of 169 DU, holds a dusk zenith near 10,800 K. The same thinning is visible in the ozone-hole years. A polar-spring column of 130 DU leaves the dusk zenith pale blue, near 7,800 K, against 12,500 K today. Beyond the ozone story: Hadean sunsets were not events at the horizon but a slow dimming of the whole glowing dome; hazy Archean sunsets were the longest and most saturated in Earth's history, vermilion at the horizon and amber overhead; Snowball sunsets were brief and golden; the sunsets of the impact winter did not exist as color at all; and the reddest, most theatrical skies of the Phanerozoic belong to volcanic years like 1815 and 1883.

3.5 The atmosphere from space

To show latitude and altitude at once, these renderings look at Earth from space with the Sun behind the viewer at the equinox, so the solar zenith angle at every point equals its latitude (equator at the center line, poles at top and bottom). The shell around the disk is the atmosphere's limb: the color of light scattered toward the viewer along a tangent ray at each altitude, computed with the same model. The atmosphere is drawn 30 times too thick, so 100 km spans half an Earth radius; a real limb is a hairline. The disk shows the planet's reflected color over a dark ocean (over ice for Snowball Earth), with limb brightness compressed so faint upper layers remain visible.

Table 3: Summary of sky colors by epoch (mid-latitude noon unless stated).

Epoch	Noon zenith, mid-latitude	Horizon	Sunset
4.55 Ga, proto-Earth	Pale blue-white under 3 bar of CO_2 and dust	Milky	Orange; a Sun larger and redder than today's
4.5 Ga, 30-bar CO_2	Peach-white, 3,300 K, shadowless	Same as zenith	Whole dome dims to orange; Sun rarely visible
4.0 Ga, 0.5-bar CO_2	Pale milky blue, 10,500 K	Milk white	Deep red Sun; cream twilight (no ozone)
3.8 Ga, clear Archean	Blue, 13,000 K	White	Modern-like, then yellow-beige twilight zenith
2.7 Ga, thin haze	Pastel milk-blue, 8,600 K	Warm white, 7,100 K	Orange twilight all the way to the zenith
2.7 Ga, thick haze	Cream-tan, 5,600 K	Cream-tan	Vermilion horizon, amber sky, long orange dusk
2.7 Ga, very thick haze	Saturated apricot, 3,900 K	Same as zenith	Sun lost in the haze before it sets; orange dusk
2.2 Ga, post-GOE	Blue, 13,300 K	White	Pale blue-gray twilight zenith (∼7,100 K); deep blue dusk comes later
700 Ma, Snowball	Deep blue, 16,200 K	Blue-white, stays blue	Golden, pale
466 Ma, meteor storm	Blue, 14,200 K	White, 8,400 K	Modern-like; at night a hundred times today's fireballs and a bright zodiacal band
300 Ma, 33% O_2	Blue, 13,500 K	White, slightly brighter	Modern-like; smoky more often
66 Ma, impact winter	Dim amber-beige, 4,600 K	Beige	None; dome fades to brown
1.78 Ma, ζ Oph supernova	Same blue as today	White	Modern; a magnitude −11.6 supernova low in the south
342 ka, Geminga supernova	Same blue as today	White	Modern; a magnitude −11 supernova in Orion that casts shadows
1815, volcanic year	Milky blue-white, 9,200 K	Slightly bluer than zenith	Salmon Sun, pink-lavender afterglow
Ozone-hole spring	Blue, still	White	Pale blue dusk zenith, ∼7,800 K
Today, clean	Blue, 15,000 K	White, 8,400 K	Orange horizon, pink antisolar; twilight zenith ∼12,500 K
Today, polluted city	Pale gray-blue, 8,200 K	Slightly bluer than zenith, 8,900 K	Sun lost in gray murk before the horizon
2100, megaconstellation	Same blue as today	White	Thousands of bright datacenters along the dusk terminator, plus a few hundred communications satellites



Modern **3.8 Ga, clear Archean** **2.7 Ga, hazy Archean** **4.5 Ga, 30-bar CO_2**

700 Ma, Snowball **300 Ma, Carboniferous** **66 Ma, impact winter** **1815, volcanic year**

Figure 2: Volumetric renderings of the atmosphere at eight epochs. Equinox geometry with the Sun behind the viewer; the shell is the limb color along tangent rays, drawn 30 times too thick (100 km spans half an Earth radius); the disk is the planet's reflected color over ocean (ice for Snowball Earth). Limb brightness is compressed so faint upper layers remain visible.

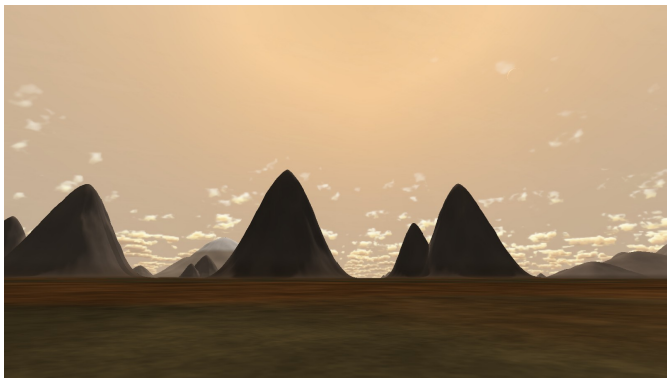


Figure 3: Noon months after the Chicxulub impact, at 66 Ma: soot and sulfate turn the sky a dim, uniform amber-beige with no blue at all, about a fifth as bright as a clear sky, with almost no cloud.

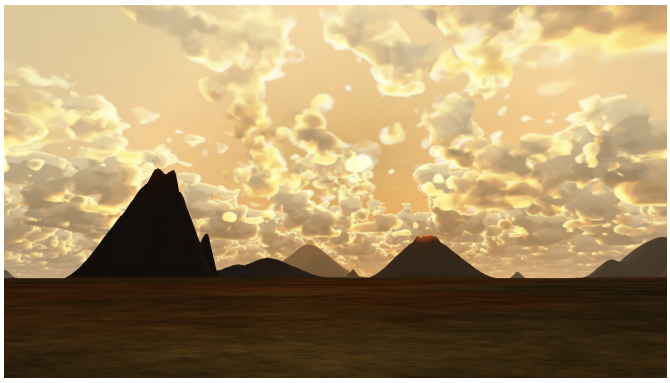


Figure 4: An afternoon at 2.7 Ga under the thick organic haze, as the site's first-person view draws it: a pale orange dome with almost no difference between zenith and horizon, the Sun a soft disk, and sparse cumulus.

Modern, clean air. Modern Earth for reference: white-blue troposphere grading to a deep blue stratosphere and fading to black, brightest at the equator and darker toward the poles.

Early Archean, ∼3.8 Ga. Clear early Archean. Structurally modern: white-blue near the surface, deep blue above, fading to black. With a little more CO_2 and no ozone the shell is marginally paler than today's and the disk very slightly whiter.

Neoarchean, ∼2.7 Ga (thick haze). Hazy Neoarchean. The organic haze colors the entire limb tan: every tangent ray below ∼60 km crosses the layer (centered ∼45 km) twice. Only above it does the shell turn a faint, dusty blue. The disk is the pale orange dot, though from space it is a subtle cream-orange rather than a vivid one.

Early Hadean, ∼4.50 Ga. 30-bar CO_2 Hadean. The shell is an opaque, luminous white-peach almost all the way up, because tangent rays at any tangent height below ∼60 km pass through many optical depths of CO_2; only the outermost fringe thins to blue. The disk is bright and nearly white: a cloudless, Rayleigh-only Venus.

Snowball Earth, ∼700 Ma. Snowball Earth. The bluest shell in the set above a blazing white disk of ice. The polar limb stays bright because the frozen surface reflects so much light back into the atmosphere.

Carboniferous, ∼300 Ma. Carboniferous, 33% O_2. The extra oxygen raises total pressure by about a tenth, so the limb is marginally thicker and brighter than today's, but to the eye it is indistinguishable from modern Earth: once ozone and clean air are in place, the sky from space has been the same pale blue dot for two billion years.

K-Pg impact winter, 66 Ma. Impact winter. A dim brown-gray band from the soot and sulfate layer occupies the lower and middle shell; above ∼40 km the untouched upper atmosphere is still faintly blue. The disk is a dull gray-lilac, and dark at high latitudes.

Volcanic year (Tambora 1815 / Toba-lite). Volcanic year. A milky white sulfate band near 20 km sits between the white-blue troposphere and the deep blue upper shell. The disk is brighter and whiter than clean modern Earth.

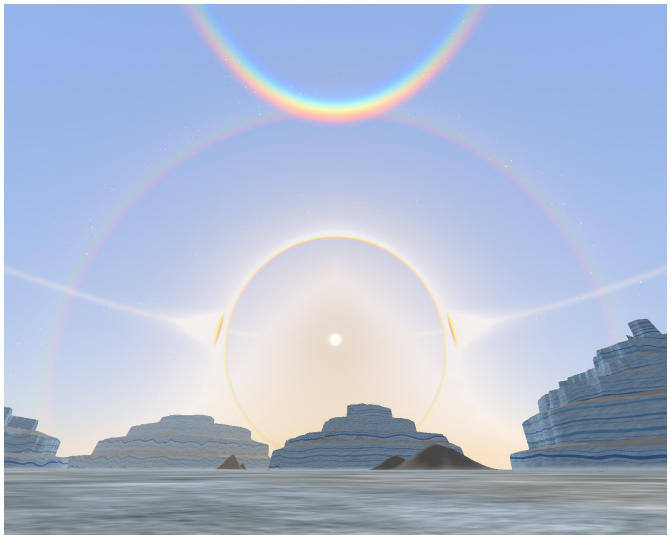


Figure 5: A morning on Snowball Earth at 700 Ma, from mid-latitudes, the Sun 22° up through clear air full of diamond dust: the 22° halo with its tangent arcs hugging its top and bottom, the sun dogs just outside it, the faint 46° halo, the circumzenithal arc high overhead, and the parhelic circle running level through the Sun. Each arc is red on its side toward the Sun.

3.6 Clouds

Clouds are not part of the radiative-transfer model; the first-person view draws them as scenery, ray-marched through a volume and lit by the model's sky, the Sun and the Moon, and by a supernova, a fireball or city light when there is one. Their regimes are illustrative choices for each era rather than results: a near-overcast stratiform deck under the steamy Hadean CO_2 (88% cover at 4.5 Ga), scattered cumulus in the Archean thinning as the haze thickens (from 48% at 3.8 Ga to 16% under the thickest haze), sparse thin cloud over the frozen Snowball (22%), towering convection in the warm Carboniferous (82%), almost none in the impact winter (8%), and more cirrus in the volcanic year. The weather changes over days and the clouds through each day: cumulus eras build to an afternoon peak and stratiform ones burn off by midday. After sunset the clouds stay lit from below until the Earth's shadow climbs past them.

3.7 Crepuscular rays

Where the Sun shines from behind a cloud, beams fan out across the sky and the sky between them darkens: crepuscular rays, the cloud's shadow in the sunlit air. Only the part of each direction's light that is sunlight scattered once in the lower air can be shadowed, so the page computes that share from each epoch's own air, its gas, aerosol, dust, haze, sulfate and soot layers with their heights, for every Sun height and direction (*shafts.py*); along each line of sight the stretches whose path to the Sun passes through cloud lose their share. The rays are strongest where the scattering sits low, and weakest in the 30-bar Hadean, whose light is multiply scattered and comes from everywhere at once. The Moon makes the same rays through gaps in cloud, and so does a bright fireball.

3.8 Ice halos

In clear, very cold air the sky fills with diamond dust, tiny hexagonal plates and columns of ice drifting near the ground. Snowball Earth's air would have been full of them at every latitude (Figure 5). Today they are common on the Antarctic plateau and in Arctic winter, so they are drawn at 75° in the modern epochs and in the other cold epochs (2.2 Ga at the end of the Huronian glaciations, the Late Ordovician and Late Paleozoic ice ages, and the Pleistocene), and faintly at 45° at 342 ka, a glacial maximum.

The halos are traced from the crystals' geometry with the refractive index of ice at 450, 550 and 650 nm (Warren & Brandt 2008), so each arc carries its own dispersion, red toward the Sun: the 22° and 46° halos, the upper and lower tangent arcs, the sun dogs (which move out with the Sun's height and vanish above 61°), the circumzenithal arc (only while the Sun is below 32°), the parhelic circle with its 120° parhelia, a sun pillar for a low Sun, and the diffraction aureole. The Moon makes the same, faintly. Their brightness, set against the clean noon sky, is an estimate of a good display; the air in front of them is each epoch's.

4 The Sun

4.1 The young Sun

The Sun was about 30% fainter when the Earth formed and has brightened since, as hydrogen burning has raised the density and temperature of its core. We follow the 1 $M_\odot$ track of Baraffe et al. (2015): 68% of today's light and 5,530 K at 4.5 Ga, 80% and 5,660 K at 2.7 Ga, 94% at 700 Ma, today's 5,772 K by the Cenozoic. Its radius follows from the luminosity and temperature: 0.90 of today's at 4.5 Ga, 0.93 at 2.7 Ga, 0.98 at 700 Ma, so the young Sun's disk was smaller as well as dimmer, and the closer Moon covered it easily. The 4.55 Ga Sun is different. Seventeen million years old, it is still contracting onto the main sequence: cooler, 4,614 K, at 47% of today's light but 7% larger than today's, an orange-white disk. The color change is small, two or three hundred kelvin for most of the record, and swamped by the air's; the brightness change is not, and it is why the early skies are drawn dimmer. The Earth–Sun distance varies through the year with today's eccentricity, so the Sun is 3.4% brighter at perihelion in January.

4.2 Sunspots

The Sun's disk carries its spots and faculae, drawn from a map of the whole photosphere made for each epoch and day and turned with the Sun's rotation. How spotted the Sun was follows its age. Young solar analogs are spotted over 1 to 10% of their surface, the coverage falling roughly as the inverse square root of age (Morris 2020) and much faster once the rotation period passes about 15 days (Nichols-Fleming & Blackman 2020). Stars more active than Rossby number one, the Sun before about 2.5 Gyr, dim when they are active: dark spots dominate their variability. Older, quieter stars like today's Sun brighten with activity, dominated by bright faculae (Shapiro et al. 2014; Reinhold et al. 2019), because spot area grows faster with activity than facular area does (Nemec et al. 2022). The Sun's rotation follows the slow-rotator sequence it most likely took (Gallet & Bouvier 2013): 3.1 days at 4.5 Ga, 16 at 2.7 Ga, 23 at 700 Ma.

Fast rotators carry spots at high latitudes and large cool polar caps, as Doppler images of young solar analogs show (Strassmeier 2009; Barnes et al. 2005), so the Hadean and early Archean Suns have their spots between 10° and 75° and caps round the poles. A regular cycle sets in at a few hundred million years (Katsova 2020); from 2.7 Ga the spots emerge in a butterfly pattern, from mid-latitudes toward the equator over each cycle (Hathaway 2011), on cycles lengthening from 8 to 11 years. On Gregorian dates the modern epochs follow the real cycles, each cycle's sunspot number from the SILSO record with the shape of Hathaway, Wilson & Reichmann (1994): the 1815 volcanic year falls in the Dalton Minimum and dates in the Maunder Minimum are all but spotless. Beyond the record a sixth of the time is a grand minimum, as over the Holocene (Usoskin, Solanki & Kovaltsov 2007).

Each active region is a bipolar group: a large, round leading spot ahead in the rotation and nearer the equator (Joy's law), a following polarity broken into smaller spots, often with partial penumbrae, and pores between. Peak areas are lognormal (Baumann & Solanki 2005); a group grows in days and decays linearly, faster the smaller it is, and drifts with the latitude's rotation (Snodgrass 1983). The spots have the structure seen in white light (Solanki 2003; Borrero & Ichimoto 2011): an umbra of 3,700 to 4,600 K, darker the larger it is, sprinkled with umbral dots and crossed in the larger spots by granular light bridges; a penumbra of radial filaments 150 to 650 K below the photosphere, with bright grains at their inner ends and a ragged, comb-like outer edge; and round the region, faculae that show only toward the limb, cleared from a moat round each mature spot, over the faint network of the supergranulation. Intensities are Planck ratios at the color channels' wavelengths, so spots are redder as well as darker. The disk is drawn as a white-light photograph is shown, with umbrae near-black and penumbrae mid-gray. Zoomed further, past the map's resolution of about 1,000 km, the photosphere's own fine structure takes over (*The Sun close up*).

A spot about an arcminute across, some 500 millionths of the hemisphere, is the smallest the eye can see unaided through a haze or a filter (Vaquero 2007). Today one is on the disk about one day in 25, mostly near solar maximum; at 2.2 Ga every other day; in the Archean and Hadean always, about 30 at once at 4.5 Ga. Spots then take about 3% off the Sun's light, and they gather toward the poles, darkening the top and bottom of the disk. The young Sun's shape does not change visibly: rotating in 3 days flattens it by only 0.05%, and even a 1-day rotation by 0.5%, 10 arcseconds off the width of a 32-arcminute disk.

4.3 The Sun close up

Zoomed in, down to a field of view of 14 arcseconds, the photosphere is drawn as high-resolution solar telescopes show it: granulation, convection cells about 1,300 km across, bright where hot gas rises and dark in the narrow lanes where it sinks, living about eight minutes, a fifth of the larger ones splitting from a dark knot late in life (exploding granules); magnetic bright points a couple of hundred kilometers across in the lanes inside faculae; penumbral filaments a few thousand kilometers long; umbral dots about 1,000 K hotter than the umbra around them; and toward the limb the supergranular network, cells some 30,000 km across outlined by brightenings. Spot and plage edges are ragged down to the pixel. This fine structure is the same in every epoch; only the spots and faculae change with the Sun's age (Figure 6).

In the first-person view a Sun near full daylight is not drawn as a disk at all but as the eye sees it: a blinding white blank tinted the epoch's color, in a bloom that whitens the sky around it. Zooming in, as through a filter, dims it to the real disk, and the glare eases off as the Sun sets or is eclipsed, until the corona and Baily's beads take over.

4.4 The setting Sun

Near the horizon the Sun is drawn as twenty images, one per 20 nm band from 385 to 765 nm, each lifted by its own refraction (the dispersion of air, Edlén 1966) and dimmed by its own optical depth along the grazing path through the epoch's model atmosphere. At the horizon the green and orange images of today's Sun are only about 6 arcseconds apart, so the colors separate only in the last sliver: the green flash is whichever band survives the extinction last, and the cleaner the air the shorter that wavelength. Along the horizontal path, the model's air still passes a twentieth of its peak sunlight at 550 nm in the clean, dry Snowball epoch, the best flashes in the record, and at 630 nm in today's clean air; under the Neoarchean haze, the impact-winter soot, a volcanic veil or a polluted city nothing shorter than 650 to 730 nm gets through, and there is no flash at all.

How far the air lifts the Sun follows its refractivity, the sum over its gases' partial pressures. Today's horizon refraction of about 34 arcminutes is 1.8 times larger at 4.0 Ga (half a bar of CO_2 on a bar of N_2), 1.11 times in the oxygen-rich Carboniferous and 0.83 times in Snowball Earth's thinner air, so the Sun sets later or earlier by a minute or two. The 30-bar Hadean air is 47 times as refractive as today's: a horizontal ray bends faster than the ground curves and is trapped (ducting), but its Sun is lost in Rayleigh scattering long before it reaches the horizon, so we hold the drawn bending to three times today's.

Each evening also has its own layering of the low air, set by the date so that a given evening always sets the same way. Thin temperature inversions step the disk's edge, show a strip of it twice (once inverted) where the rays cross, and draw a dark line across it where trapped rays run through the densest air; over warm water the Sun meets its own inverted image in an inferior mirage, the "omega" sunset. The cold surface of Snowball Earth makes inversions common and inferior mirages rare, and the warm oceans of the Hadean and Archean

Table 4: Sunspots by epoch, from the site's spot model. Coverage is the share of the surface in spots and their penumbrae: the mean, and the 5–95% range over the cycle or the young Sun's irregular swings. A naked-eye spot is one over about 500 millionths of the hemisphere, an arcminute across, on the visible disk; the modern row is 1996–2026 on the real cycles. Rotation is the equator's sidereal period.

Epoch	Sun's age	Rotation	Spot coverage	Spots emerge at	Cycle	Days with a naked-eye spot
4.55 Ga proto-Earth	0.017 Gyr	1.6 d	10%	25–80°; polar caps hold 45% of the spotted area	none	all
4.5 Ga Hadean	0.07 Gyr	3.1 d	9% (7–11%)	22–75°; polar caps above 72°	none	all (about 30 at once)
4.0 Ga Hadean	0.57 Gyr	9.0 d	4% (2.9–5.3%)	15–65°; caps above 81°	about 5 yr, shallow	all (about 14)
3.8 Ga Archean	0.77 Gyr	10.4 d	3% (1.9–4.3%)	10–60°; caps above 85°	about 6 yr	all (about 11)
2.7 Ga Archean	1.87 Gyr	16 d	1% (0.3–1.8%)	from 40° toward the equator	8 yr	83%
2.2 Ga post-GOE	2.37 Gyr	18 d	0.5% (0.1–1.0%)	from 34°	9 yr	50%
700 Ma Snowball	3.87 Gyr	23 d	0.25% (0.02–0.5%)	from 30°	10.5 yr	20%
466 Ma Ordovician	4.10 Gyr	24 d	0.20% (0.01–0.4%)	from 29°	10.8 yr	18%
300 Ma Carboniferous	4.27 Gyr	25 d	0.17% (0.01–0.4%)	from 28°	11 yr	12%
66 Ma impact winter	4.50 Gyr	25 d	0.14% (0.01–0.3%)	from 28°	11 yr	8%
1815, today and 2100	4.57 Gyr	25.4 d	0.09% (0.01–0.19%)	from 28°	the real cycles	4%

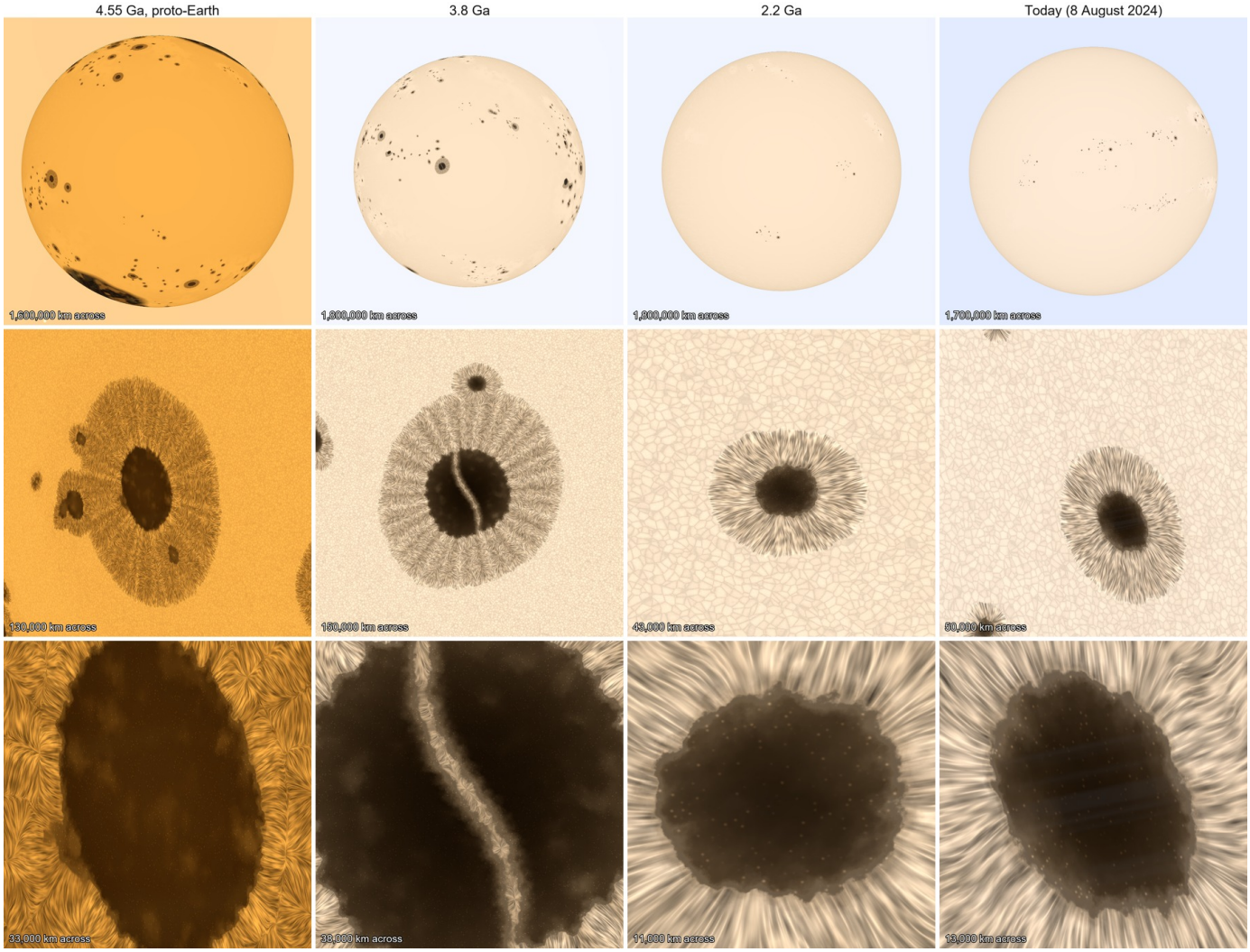


Figure 6: The Sun's surface at four epochs, as the site's telescope view draws it at noon through each epoch's air (sun_figs.py). Top: the whole disk. Middle: its largest spot group. Bottom: that group's largest umbra. Widths are in km on the Sun, since the site draws the Sun at four times its size. The 17-million-year-old Sun of 4.55 Ga, orange through the proto-Earth's dusty air, is spotted over a tenth of its surface, its spots at high latitudes and a dark cap over the pole; at 3.8 Ga spots are on the disk every day; by 2.2 Ga a few groups keep to the butterfly's latitudes; today's disk is the real cycle 25 near its maximum. Close up the structure is the same in every epoch: granulation and its network around the spots, penumbral filaments, umbral dots, and in the 3.8 Ga spot a light bridge of granules.

the reverse. These mirage statistics are illustrative. The disk is also darker and redder toward its edge (limb darkening, Hestroffer & Magnan 1998): the air only multiplies the Sun's light, so the darkening is the same under every sky.

Two of these effects are drawn larger than life. The disk itself is drawn at four times its angular size, as the Moon is, so that it shows at all on a screen, and the bands' separation by refraction is enlarged with it, four times high in the sky and up to 24 times at the horizon, where a real flash is a sliver of a few arcseconds. And the limb darkening is drawn at a fifth of its true strength: on a disk drawn near white, the real contrast reads as a dark rim. The spectrum readout keeps the full law.

4.5 Solar eclipses

When the Moon covers part of the Sun, all the sunlit sky dims by the fraction of the disk still showing. In totality the sky is lit from beyond the Moon's shadow, tens to a hundred kilometers away, where the Sun is still partly up: the epoch's own twilight sky with a sunset glow all round the horizon, a thousandth of the day sky, as measured (Sharp, Lloyd & Silverman 1966). The Hadean and hazy Archean totality skies are therefore that era's dusk, peach or amber rather than the deep blue of a modern eclipse.

The corona follows the Sun's age. A younger Sun spun faster and was far more active: its X-ray output goes as age to the -1.5 (Güdel, Guinan & Skinner 1997), about 530 times today's at 4.5 Ga, and its coronal temperature with it, 7.7 MK against 1.5 MK. Today's corona (Baumbach 1937: electron-scattered K light near the limb, dust-scattered F light beyond about 2.3 solar radii) is scaled accordingly: a denser, more extended K corona, an F corona that follows the era's zodiacal dust, stronger coronal lines (the green Fe XIV and red Fe X, joined by yellow Ca XV in the hot young corona) and a brighter pink chromosphere. These scalings are estimates. The young Sun was also smaller (0.90 of today's radius at 4.5 Ga) while the Moon was closer, so early eclipses hide the inner corona: at 3.8 Ga the Moon looks 1.5 times the Sun's width, and from 2.2 Ga back every central eclipse is total. Through the 30-bar Hadean air, at 5.8 magnitudes of extinction per airmass, behind a Moon eight times the Sun's width, the corona hardly shows: only its outer part clears the Moon's limb, a faint glow lost in the molten Moon's own aureole. The proto-Earth of 4.55 Ga has no Moon, and so no eclipses; its Sun, 17 million years old, has the saturated corona of a star that young, at about 9 MK.

The eclipse itself is drawn from the Moon's real profile: at second and third contact the last sunlight breaks into Baily's beads where the lunar limb's valleys, from the Laser Altimeter's heights, let the photosphere through, and a diamond ring. Like the disks, the eclipses are enlarged: the Sun and Moon are drawn at four times their size, and eclipses are found and timed with the enlarged disks, so the page's partial phases last longer, and its eclipses happen at more places, than real ones. For each new Moon the page stands where parallax brings the Moon closest to the Sun, the best seat for that eclipse.

5 The Moon

5.1 The Moon's orbit through time

The Moon raises tides on the Earth, and the tidal bulge, carried ahead of it by the Earth's spin, pulls it forward: the Moon has been receding ever since it formed, today by 3.8 cm a year, while the Earth's day lengthens. Run back, the Moon was closer and bigger, its month shorter, and the day shorter too. We take the distance from Farhat et al. (2022), whose tidal model fits both the present recession and the geological record of tidal rhythmites, back to 3.2 Ga, where Eulenfeld & Heubeck (2023) find 70% of today's distance in the Moodies Group rhythmites; older ages extend the long-term drift. At 4.5 Ga, some ten million years after the giant impact, the Moon is still at about 8 Earth radii: Korenaga (2023) finds that a Moon whose magma ocean was still solidifying raised little tide in itself and had receded only to 7 to 9 Earth radii by the time the ocean froze. (Table 5 gives the numbers by epoch.)

The day follows from the distance by holding the Earth–Moon angular momentum fixed: a closer Moon means a faster-spinning Earth. This reproduces the day lengths Farhat et al. (2022) give for their distances to within a tenth of an hour, 21.9 hours at 700 Ma and 15.6 hours at 2.7 Ga, and at 4.5 Ga it gives 5.9 hours. The year is the same length in seconds in every epoch, since the Earth's orbit is set by the Sun's mass, which has barely changed; so a year held 400 days in Snowball Earth, 562 at 2.7 Ga and 1,486 at 4.5 Ga. The month, by Kepler's third law, was 22.6 of today's days at 2.2 Ga and 15.3 at 4.0 Ga, still 28 to 31 of each epoch's own shorter days; at 2.7 Ga a year held about eighteen months, as tidal rhythmites suggest. At 4.5 Ga the Moon went round in 32 hours, five and a half of that epoch's days. The proto-Earth of 4.55 Ga has no Moon; how fast it turned before the impact cannot be known, and its 12-hour day is a guess.

The orbit's tilt to the ecliptic also changed. We carry today's 5.15° back along these distances with the semi-analytic tidal model of Ćuk et al. (2016): the Earth's tides tilt the orbit less as the Moon recedes, and tides raised in the Moon itself damp the tilt, most strongly near 33 Earth radii, where the Moon's own obliquity swings large. The orbit comes out a little steeper in the past, 5.5° at 2.7 Ga and 5.8° at 4.0 Ga; Ćuk et al. argue these are lower bounds, and the 17° to 30° and more that such models give for the young Moon belong before 33 Earth radii, older than any epoch here but 4.5 Ga. Inside the Laplace radius the orbit precesses about the Earth's equator rather than the ecliptic (Tremaine, Touma & Namouni 2009, eq. 24): 8.4 Earth radii for the present Earth. A fluid Earth spinning in 5.9 hours carries a larger bulge, and the radius moves out by $(24/5.9)^{2/5}$, to about 15 Earth radii, so the 4.5 Ga Moon is drawn in the plane of the equator. The orbit was more eccentric too, growing from today's 0.055 to about 0.1 at 4.0 Ga (Daher et al. 2021), after it had been through the evection resonance. In the Ćuk & Stewart (2012) example that resonance holds the eccentricity near 0.45 from about 9,000 to 70,000 years, and by 4.5 Ga tides have damped the orbit nearly round again. With the eccentricity the Moon rocks east and west each month by about twice the eccentricity in radians, $\pm 6.3°$ today and $\pm 12°$ at 4.0 Ga, and north and south by its obliquity, 6.7° today and 21° to 23° at 3.8 and 4.0 Ga, as it nears the Cassini-state transition; the page draws the globe on these axes, so the Moon visibly librates.

The consequences show in the sky. At 3.8 Ga the Moon is one and a half times today's width and more than twice as bright when full. At 4.5 Ga it is 4° across, seven and a half times today's width and 57 times its area, and so close that it is placed as seen from the ground: it shifts 7° against the stars between the horizon and the zenith. It crosses the sky in hours, rising and setting several times a day. In the plane of the equator, it eclipses the Sun at every new Moon for weeks around each equinox, somewhere on the Earth, and passes through the Earth's shadow, 14° across at its distance, at every full Moon.

5.2 The Moon's face

The Moon's face is the Lunar Reconnaissance Orbiter's: the Wide Angle Camera's color mosaic, toned to match a full-Moon photograph, and the Laser Altimeter's heights (NASA's CGI Moon Kit). The heights light the slopes, cast shadows toward the terminator, and shape the limb, so a zoomed crescent shows a ragged terminator and crater shadows, and the edge of the Moon in an eclipse breaks the last sunlight into Baily's beads. Five globes cover the Moon's history (Figure 10). Tycho, about 109 million years old (Stöffler & Ryder 2001, from the exposure ages of Arvidson et al. 1976), is taken away with its rays for the epochs older than 109 Ma. Before about 3.9 Ga the maria had not been flooded: basalts filled the great basins from about 3.9 Ga and most of them by 3.3 Ga (Hiesinger et al. 2011), so each mare darkens over 150 million years from an onset just before its oldest dated basalt. At 3.8 Ga Tranquillitatis is two-thirds dark, Serenitatis



Figure 7: A total solar eclipse in five epochs, as the site's first-person view draws it: in each, the first total eclipse after October 2026 on the page's calendar with the Sun at least 25° up (today's is the eclipse of 30 March 2033), cloud cleared. Left to right: the partial phase, 70 percent of the way from first to second contact; the diamond ring, with a sliver of photosphere half a percent of the Sun's width left; mid-totality; and mid-totality in a 75° view with the horizon. The close-ups share one scale (14° high), with the Sun and Moon at four times their size as on the site, so the growing Moon shows, from about the Sun's width today to 1.15 times it at 2.2 Ga and 1.55 times at 4.0 Ga, where it hides the inner corona; each frame is exposed as the site's view adapts to its light. Totality's sky is the epoch's twilight: slate blue-grey with a sunset rim on the horizon today, at 300 Ma and at 2.2 Ga, amber over an orange horizon under the 2.7 Ga haze, which also dims the corona, and the hot young Sun's corona at 4.0 Ga is far larger and brighter. The young Sun's spots show on the crescents at 2.7 and 4.0 Ga. At 4.5 Ga the Moon is eight times the Sun's width and the corona hardly clears it.

Table 5: The Moon by epoch, as the site draws it. Distance in Earth radii (Farhat et al. 2022; Eulenfeld & Heubeck 2023; Korenaga 2023 at 4.5 Ga); width and full-Moon light relative to today's; the day from the Earth–Moon angular momentum held fixed; the month from new Moon to new Moon in today's days, and in brackets in the epoch's own; the orbit's tilt from the tidal model of Ćuk et al. (2016), in the plane of the equator at 4.5 Ga; the eccentricity after Daher et al. (2021). The 4.55 Ga proto-Earth has no Moon.

Epoch	Distance ($R_\oplus$)	Width	Full-Moon light	Day	Days a year	Month (own days)	Tilt to ecliptic	Eccentricity
Today	60.1	1.00×	1.00×	24.0 h	365	29.5 d (29.5)	5.15°	0.055
66 Ma	59.9	1.00×	1.01×	23.8 h	368	29.4 d (29.6)	5.15°	0.056
300 Ma	58.6	1.03×	1.05×	22.6 h	388	28.3 d (30.0)	5.18°	0.059
466 Ma	58.2	1.03×	1.07×	22.3 h	393	28.0 d (30.1)	5.19°	0.060
700 Ma	57.7	1.04×	1.09×	21.9 h	400	27.6 d (30.3)	5.20°	0.061
2.2 Ga	51.0	1.18×	1.39×	17.4 h	504	22.6 d (31.2)	5.37°	0.078
2.7 Ga	47.6	1.26×	1.60×	15.6 h	562	20.3 d (31.2)	5.46°	0.087
3.8 Ga	40.4	1.49×	2.22×	12.8 h	685	15.7 d (29.4)	5.75°	0.105
4.0 Ga	39.8	1.51×	2.28×	12.6 h	696	15.3 d (29.2)	5.80°	0.106
4.5 Ga	8.0	7.52×	57×	5.9 h	1,486	1.3 d (5.4)	equator	0.010

and Nectaris a third, and Imbrium and Procellarum are still bright: the familiar face is not there yet. At 4.0 Ga the near side is bright highland from limb to limb. At 3.8 Ga the eruptions themselves show: one to three fissure eruptions a lunation glow on the night side, a few to a hundred square kilometers of lava at about 1,400 K, as bright per area as the sunlit Moon and far redder, magnitude +3 to −1 as seen from the Earth.

The Moon of 4.5 Ga is not yet solid. Its magma ocean is open at the top, some ten million years after the impact: rafts of dark quench crust and the first pale anorthosite, floating a few hundred meters to a kilometre and a half above the melt, broken by cracks and lava lakes at 1,400 to 1,600 K in four generations of floes, each a network of cracks within the last. The lava shines about as brightly per area as the sunlit Moon, so the disk glows orange by day and night alike, also inside the Earth's shadow, and the air around it carries a red aureole. The map and its relief are made up; that the young Moon looked like this is an inference from its magma ocean, not an observation.

The night side is drawn at a flat 6% of the lit side's brightness rather than with a modeled earthshine, which the spectrum readout gives instead: about 2×10^{-4} of the day side when the Earth is full from the Moon, and bluer.

5.3 Moonlight

Moonlight is the same sky model with the Moon in place of the Sun, scaled by the lunar phase (the Allen phase law with the opposition surge, as in Krisciunas & Schaefer 1991) and by the inverse square of the Moon's distance. It stays nearly the same in color from epoch to epoch but not in brightness: the full Moon lit the sky 2.2 times as brightly at 3.8 Ga and 57 times at 4.5 Ga, where it lit a 30-bar sky that, like its day sky, was a featureless peach glow.

5.4 Lunar eclipses

In a lunar eclipse the Moon passes through the Earth's shadow, which is lit only by sunlight bent into it through the Earth's limb: every sunrise and sunset on Earth at once. Its color and brightness are a measurement of the whole atmosphere at the terminator, which is why eclipses darkened after Tambora and El Chichón (Stothers 2005; Keen 1983), why the eclipse of August 2008 carried the imprint of the small Kasatochi eruption (García Muñoz et al. 2011), and why the eclipsed Moon's spectrum is Earth's transmission spectrum as an exoplanet would show it (Pallé et al. 2009). The same physics, run on each epoch's air, gives the eclipse of every era.

The model (*lunar_eclipse.py*) traces rays from the Sun through the epoch's atmosphere at tangent heights from the ground to 120 km, bending with the refractivity of its gas mix and accumulating the optical depth of every component (gas, ozone, aerosol, sulfate, haze, soot) along the whole path, with today's cloud tops cutting the lowest rays. Seen from a point on the Moon, the Earth's limb at each position angle and height shows the Sun's limb-darkened disk wherever the ray's bending puts it; summing that over the ring of the limb (Link's method; Link 1969; García Muñoz et al. 2012 for a modern treatment) gives the spectrum of the light at each distance from the shadow's axis, and the direct sunlight past the top of the air gives the penumbra.

Today's umbra comes out a dull copper. A central eclipse puts the Moon 12.3 magnitudes below the full Moon, about V = −0.4, in the range observed for clear-air eclipses, with the center darkest and a paler, slightly bluish edge where ozone's Chappuis band removes the orange from the highest rays (the turquoise fringe of eclipse photographs). Before the Great Oxidation there is no ozone and that edge is orange too. A Tambora-sized sulfate veil puts the Moon near V = +11, a surface brightness of 27 mag/arcsec^2, far below the night sky: it vanishes, as it did to observers in June 1816 (Stothers 2005). The K–Pg soot and the thick Neoarchean haze remove it entirely; the thin haze leaves a dim brown-red Moon near V = +5.

Two effects are new to the deep past. The closer Moon sat nearer the Earth than the refracted light comes to a focus (about 300,000 km today, past the Moon's 384,000), so the middle of the shadow is reached only by rays skimming the ground and goes dark: at 3.8 Ga the center of the shadow is a hundred million times fainter than its edge, a dark core in a glowing ring. And the 30-bar Hadean air is so refractive that below about 19 km its rays are trapped and run round the planet, while the upper air bends light strongly enough to fill the whole umbra of a Moon some 39 Earth radii out, an even, deep red at V = −1.7; but the Moon of 4.5 Ga, only 8 Earth radii out, sits well inside the focus, and at mid-eclipse it is gone, V = +12.

The shadow's geometry follows the distances and the young Sun's size. The umbra was 3.1 Moon widths across at 4.0 Ga against 2.7 today, and with twice as many full Moons a year, total lunar eclipses came nearly three times as often, while totality itself lasted at most 105 to 109 minutes in every epoch from 4.0 Ga on (Table 6). At 4.5 Ga the Moon, inside the Laplace radius of a fast-spinning Earth, orbits in the plane of the Earth's equator, its shadow is 3.6 Moon widths across and it races through it: around each equinox every full Moon is eclipsed, some 3,900 total eclipses a century, none longer than an hour. The rate also depends on the tilt of the Moon's orbit. It was a little steeper then: integrating today's 5.145° back along the distance history with the tidal model of Ćuk et al. (2016), in which Earth's tides flatten the orbit as the Moon recedes and tides raised in the Moon damp it, gives 5.46° at 2.7 Ga and 5.80° at 4.0 Ga (a little more for a Moon stiffer or more dissipative). The much larger tilts that model and Downey, Nimmo & Matsuyama (2023) need for the young Moon belong to before the Cassini-state transition near 33 Earth radii, older than any epoch here but 4.5 Ga, whose Moon is in the equator's plane (*lunar_inclination.py*).

The ephemeris is checked against the almanac. For the total eclipses of 2022, 2025, 2026 and 2029, contact times fall within 2 to 7 minutes of the NASA catalog (Espenak & Meeus 2009) and umbral magnitudes within 0.04 (for 2029 June 26, 1.831 against 1.844); over 2000–2100 the page finds 144 umbral and 84 total eclipses, against the catalog's 142 and 85. On the page the Moon is drawn for an eye adapted to its brightest part, so the umbra looks black beside the penumbra in a partial phase and glows in totality, as it does to an observer.

6 The planets and their moons

6.1 Where the planets were

The five naked-eye planets, Uranus and Neptune are placed from JPL's approximate Keplerian elements (Standish's table, valid 1800–2050) for the date and time, light-time corrected, on the same magnitude scale as the stars (Meeus's formulae with the phase angle, Saturn's rings included when there are rings, and Mallama & Hilton 2018 for Uranus and Neptune), and dimmed by the young Sun's fainter light: 0.42 magnitudes at 4.5 Ga, 0.24 at 2.7 Ga. Any planet brighter than magnitude −6 lights the sky as the Moon does. In the historical and modern skies the positions are the real ones. Before history they cannot be known: the Solar System is chaotic, its uncertainties growing tenfold every ten million years or so (Laskar 1989), and every epoch's age is uncertain by far more than any planet's period. So each planet is put at a random place on its orbit, the same for every visit to an epoch, and moves on with the date. The orbits' shapes are today's, known to some 50 million years and statistically before.

The two oldest skies have a different Solar System. In the reading of the meteorites' bombardment ages (Edwards et al. 2024; Liu, Raymond & Jacobson 2022), the giant planets went unstable early, within the first 100 million years; the 4.55 Ga sky shows them before it, in the best starting configuration of Nesvorný & Morbidelli (2012): Jupiter at 5.6 AU and Saturn at 7.34 AU in a 3:2 resonance, a fifth giant at 9.62 AU, Uranus at 15.3 and Neptune at 20 AU, each in resonance with the next. The fifth giant, an ice giant like Neptune that Jupiter threw out of the Solar System in the instability (Nesvorný 2011), is named here Oceanus. For the 4.5 Ga sky we take the instability to have come late enough, near 4.5 Ga, to keep it; the bombardment



Figure 8: An evening at 4.5 Ga. Through 30 bar of CO_2 and a near-overcast deck, the molten newborn Moon, 8 Earth radii away and 4° across, glows orange through cracks in its first crust, with its red aureole, over a hot steaming sea.

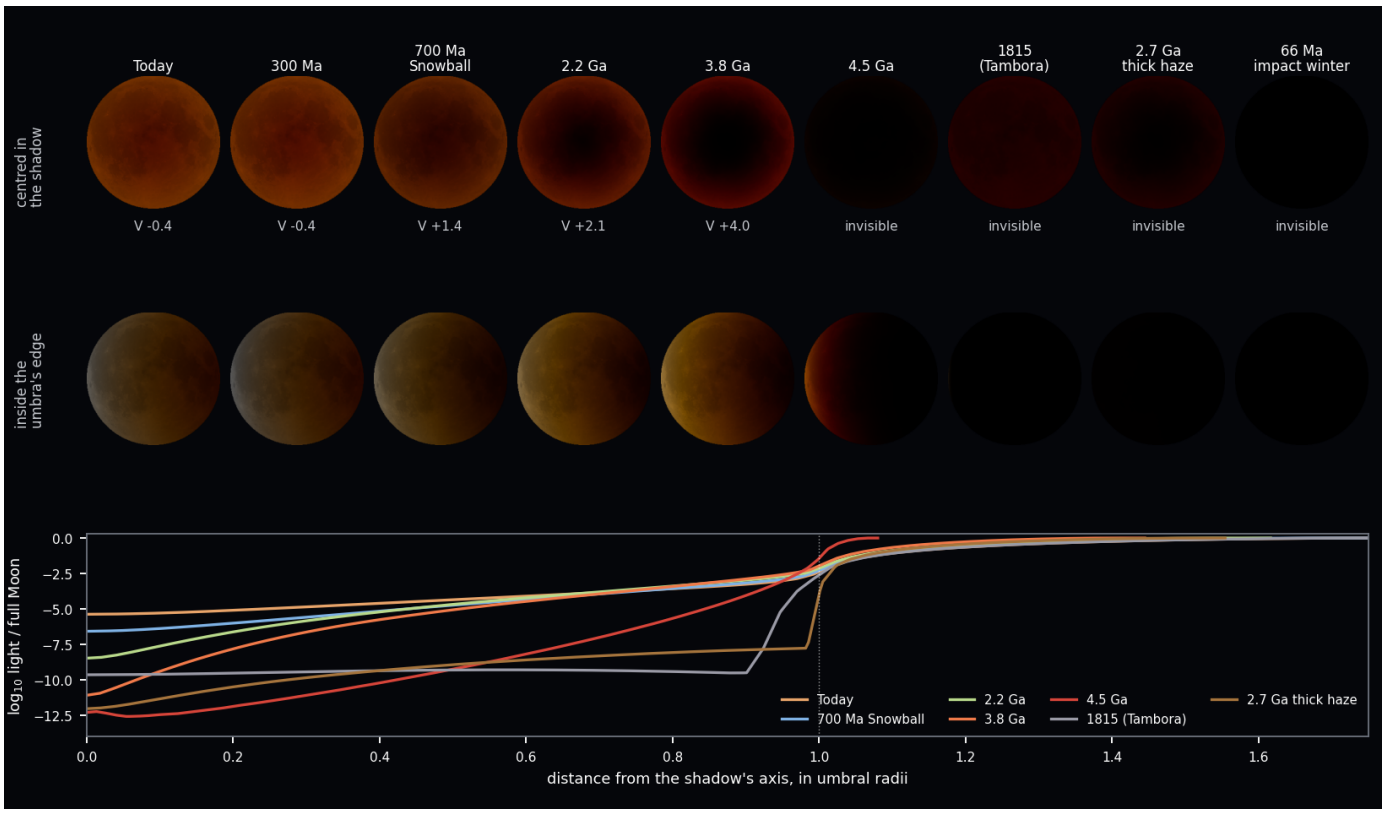


Figure 9: The Moon in the Earth's shadow at nine epochs, on today's lunar surface. Top: centered in the shadow, with its V magnitude, each drawn as the page draws it, for an eye adapted to that Moon's brightest part. Middle: just inside the umbra, its limb at 0.95 of the umbral radius, all at one exposure so their light compares: today's ozone leaves a cool gray-blue edge, the ozone-free Archean edge is orange and the Hadean's deep red, and under the Tambora veil, the thick haze and the K–Pg soot the umbra is dark to its edge. Bottom: the light across the shadow, against the full Moon, with distance in units of each epoch's umbral radius.

Table 6: Lunar eclipses by epoch. The umbra's diameter is in Moon widths at the mean distance, with Danjon's enlargement. V and the mean surface brightness are for the Moon centered in the shadow, from the model's shadow and the Moon's distance; a dark night sky is about 22 mag/arcsec^2. Counts are the page's own over a hundred years of dates (2000–2100), so they carry a few percent of noise from the eclipse cycles. Orbit tilt from lunar_inclination.py; at 4.5 Ga the Moon orbits in the plane of the Earth's equator, so its tilt to the ecliptic is the obliquity.

Epoch	Umbra (Moon widths)	V at mid-eclipse	Surface brightness (mag/arcsec^2)	Orbit tilt	Umbral / total a century	Longest totality (min)
Today	2.69	−0.4	16	5.15°	144 / 84	107
1815 (Tambora)	2.69	invisible	27	5.15°	144 / 84	107
66 Ma impact winter	2.70	invisible	—	5.15°	149 / 71	107
300 Ma	2.73	−0.4	16	5.18°	167 / 76	107
466 Ma	2.74	+0.3	16	5.19°	171 / 74	106
700 Ma Snowball	2.76	+1.4	18	5.20°	172 / 77	108
2.2 Ga	2.90	+2.1	19	5.37°	226 / 124	106
2.7 Ga thick haze	2.96	invisible	29	5.46°	286 / 137	109
3.8 Ga	3.09	+4.0	21	5.75°	420 / 211	105
4.0 Ga	3.10	−0.4	17	5.80°	438 / 218	106
4.5 Ga	3.59	invisible	33	23.44°	7080 / 3916	57

ages argue against that, as the 4.55 Ga sky's account says.

Theia, the Mars-sized body whose collision with the proto-Earth made the Moon, is a wandering planet in the 4.55 Ga sky. Its orbit is made up, as any that would meet the Earth's will do, since where it lay is not known (Branco, Machado & Raymond 2025). Every 25 years or so it passes within a few hundredths of an AU, and at its closest it reaches magnitude −8.5, half lit, brighter than any planet has been since, lighting the night. Its face is invented too: an airless, cratered, iron-rich basaltic crust (Meier, Reufer & Wieler 2014), its steam atmosphere lost to the young Sun within a few tens of millions of years (Odert et al. 2018), with eight great basins and a few young rayed craters.

The three largest asteroids are placed the same way, in every epoch: Vesta, which reaches magnitude 5.2 and the naked eye, Ceres and Pallas. Their elements are fit here to JPL Horizons over 1800–2200, with periodic terms in the mean longitude for Jupiter's pull (for Pallas, 0.9 degrees over 83 years), to a tenth or two of a degree as seen from the Sun, and their brightness follows the H, G system of Bowell et al. (1989) with JPL's values.

6.2 The planets close up

Zooming in acts as a telescope, gaining 5 magnitudes against the sky for each tenfold zoom (magnification spreads the sky's light and leaves a point a point), so faint moons and Uranus and Neptune come out, and Mercury and Venus show by day. Each planet is a disk of its true angular size and flattening, on its IAU pole and prime meridian, lit from the Sun's side, so the phases of Mercury and Venus show; its limb darkens by a Minnaert law. Mercury and Mars wear spacecraft maps, MESSENGER's color mosaic and Viking's, toned toward true color, with their relief from MESSENGER (Becker et al. 2016) and the Mars Orbiter Laser Altimeter: slopes lit by the planet's own law, and rims and peaks hiding the Sun, so a gibbous Mars shows the shadows of Tharsis and Olympus Mons near its terminator and a half-lit Mercury a ragged, cratered one. Venus is cloud, cream and featureless, bright in the crescent. The moons of Jupiter, Saturn and Uranus, and Triton, have their own maps (Galileo, Voyager and Cassini mosaics), and cast their shadows on their planets, go dark in the planets' shadows, and pass behind and in front of them. Of Uranus's five, as of Triton, Voyager 2 saw only the southern halves, lit in 1986; the rest is filled in smooth, and since Uranus's north has turned toward the Sun and the Earth, it is mostly that blank half that faces us now. Uranus has its ten narrow rings, as French et al. (2024) fit them to thirty years of stellar occultations: each the ellipse of its orbit, its periapse precessing (the ε ring's 406 km off-centre, turning once in nine months), its width growing from periapse to apoapse (the ε ring's from 20 to 97 km) and its depth thinning to match, with optical depths after Esposito et al. (1991). The particles are as dark as charcoal, so even the ε ring is only a faint thread at the deepest zoom, as it is in a real telescope; the dusty rings, a hundred-thousandth as thick, are left out. Ceres and Vesta wear Dawn's Framing Camera mosaics and stand on its shapes: each is drawn as the spheroid that fits its shape best (Ceres 482 by 446 km, matching Park et al. 2016; Vesta 281 by 226 km), with what is left of the shape, up to 23 km on Vesta, as relief that lights the slopes and also bends the outline of the disk, so Vesta's south, cut by its great basins, shows flattened and lumpy. Phobos has the Mars Express map and shape the same way, though it is never more than a few pixels. Pallas, which no spacecraft has visited, is the spheroid of the shape from Marsset et al. (2020), in its own faintly blue tint. Figure 10 shows every rocky surface the site draws.

6.3 Jupiter

Jupiter close up is drawn, not mapped: its clouds change too fast for any one map to stand for an epoch. The bands lie where the Cassini wind profile (Porco et al. 2003) puts its jets, and the clouds are a tracer stirred by eddies: each point is followed back along the streamlines of a stream function of noise, three octaves of cells three times longer along the winds than across, renewing at different times, its strength greatest in the belts and toward the poles and coming and going in storm patches some 45° long. The flow keeps area, so the belts and zones and the finer banding within them marble into curls and filaments rather than creases, and the jets' own shear draws them out along the bands. Along each boundary between a belt and a zone, Kelvin–Helmholtz billows roll up, turning the way of the shear there. Dark festoons trail from the North Equatorial Belt into the Equatorial Zone with blue-gray hot spots at their heads; dark barges sit on the belt's northern edge, and white ovals in the South South Temperate Belt. The tracer approach follows the flow-field techniques of Dziewanowski (2024).

The Great Red Spot is a vortex in the same flow: an anticyclone turning the clouds along its ellipses, faster toward its edge, with a slight inflow that draws belt and zone onto its rim as spiral arms, and its turbulent wake to the west. Its history is the observed one. It was first seen for sure in 1831 (Cassini's "permanent spot" of 1665–1713 may have been another storm), so the 1815 sky has none. Its longitude follows the measured record (Peek's for 1831–1952, JUPOS since), and its size the fits of Simon et al. (2018) from 1979, 405.0 − 0.1939 Y degrees long and 107.2 − 0.0483 Y wide, and before 1979 a steady shrinking from about 35° long in the 1880s. The fits meet in a round spot 8.4° across in 2045, and it is held there for 2100: whether it shrinks on, steadies or fades is not known. It was reddest in 1879–1882 and has deepened again since 2014. In the deep past there is no Red Spot: storms like it come and go over centuries.

6.4 Saturn and its rings

Saturn close up is drawn as Jupiter is, a tracer stirred by the same flow, with its own winds, bands and colors. The winds are Cassini's at the cloud tops in 2004–2009 (García-Melendo et al. 2011), fit by twenty Gaussian jets to 6 m/s rms in System III, which also turns the planet: the broad equatorial jet near 400 m/s, and narrower jets alternating with latitude toward the poles. The belts lie where the wind falls off poleward, as Jupiter's do, the broad pair at 18–32° and fainter, narrower ones beyond. Saturn's haze mutes all of it, so the

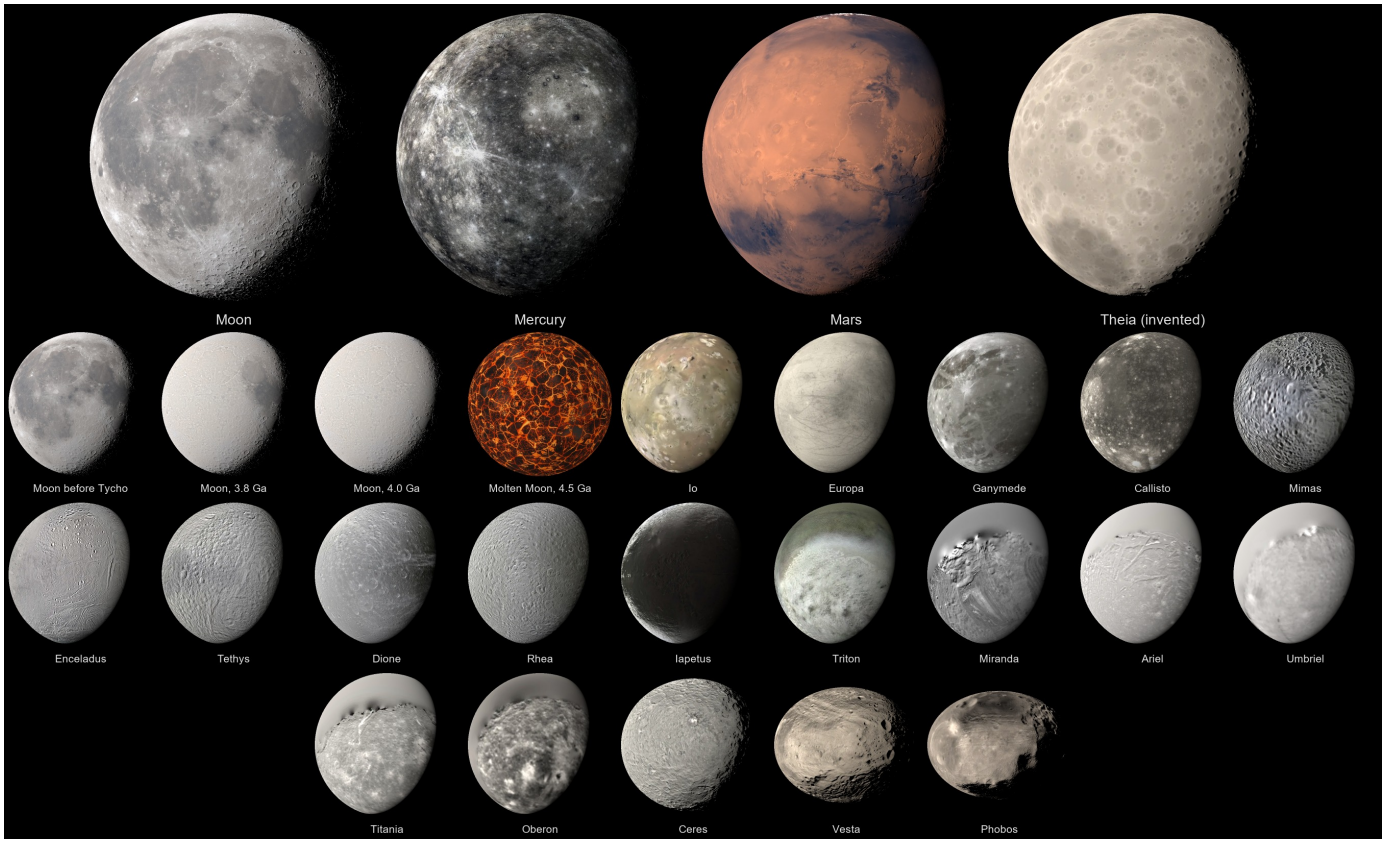


Figure 10: The rocky surfaces the site draws, each its own map on a globe lit from the left (surface_montage.py). The large panels are the four bodies drawn with heights as well, shaded by their slopes at true scale: the Moon, the Lunar Reconnaissance Orbiter's Wide Angle Camera color mosaic and Laser Altimeter heights (NASA's CGI Moon Kit); Mercury, the USGS MESSENGER MDIS color mosaic and MESSENGER heights (Becker et al. 2016); Mars, the USGS Viking color mosaic and the Mars Orbiter Laser Altimeter's gridded heights; and Theia, invented from what it was likely made of (Meier, Reufer & Wieler 2014). The Moon's older faces are today's with Tycho and its rays removed and the maria part- or un-flooded (Hiesinger et al. 2011), on the same heights; the molten Moon is a model of its magma ocean, glowing by its own light. Io, Europa, Ganymede and Callisto are the USGS Voyager and Galileo SSI mosaics; Mimas, Enceladus, Tethys, Dione, Rhea and Iapetus are Cassini color maps (NASA/JPL-Caltech/SSI/LPI, Schenk); Triton is Schenk's Voyager 2 map, its north, dark in 1989, filled in; Miranda, Ariel, Umbriel, Titania and Oberon are the USGS Voyager 2 mosaics (T. Becker), their southern halves only, shown from below. Ceres and Vesta are Dawn's Framing Camera mosaics and Phobos the Mars Express SRC mosaic (DLR, USGS), each on the spheroid the site draws it as and shaded by its shape's heights over it (Vesta's north and Ceres's far south, dark when Dawn mapped them, filled in). All are toned toward true color, and each body is shown at its own exposure, so the panels do not compare in albedo.

eddies stir about half as hard as Jupiter's, the billows turn less, and the hues vary a half to a third as much. Around the north pole the jet at 78°N meanders in a standing wave of six, the hexagon (Godfrey 1988), golden inside in the northern summer and bluish gray in the winter; both poles have a vortex with a dark eye and a bright wall, a thin ribbon wave runs along the jet at 47°N, and small bright storms rise in Storm Alley near 40°S. The winter hemisphere, out of the Sun, shows bluish at any size, as Cassini saw the north before 2009. That change lags the season by several years, and the lag is not drawn.

Saturn's rings are drawn from the Cassini radio occultation of 3 May 2005 (Marouf et al. n.d.), its optical depth at 10 km resolution from the C ring to the outer A ring, corrected to visible light in the C ring, the Cassini division and the outer A ring by the same occultation's Ka band, which sees the sub-centimetre particles the X band misses (in the outer A ring Jerousek et al. 2016 find them too), with the F ring added. Each ring is a single-scattering layer (as Cuzzi et al. 1984): lit from the Sun's side it shines by its depth and particle brightness, the A ring 0.7 of the B ring's and the C ring and Cassini division half of it and grayer; seen from the unlit side only light passing through it shows, so the B ring is black and the C ring and Cassini division glow. The rings shadow the globe, and the globe shadows the rings with its oblate outline and a penumbra the Sun's width. Near the equinoxes, when the Sun is low over the rings, they shadow the moons too: a moon passing behind them is banded by their shadow and dimmed by what they let through, Mimas by up to 4 magnitudes deep in the B ring's. Edge-on, the rings thin below a pixel.

The rings are young. How young is still argued, from tens to a few hundred million years (Iess et al. 2019; Kempf et al. 2023), and we adopt the account of Wisdom et al. (2022): an icy moon the size of Iapetus, named Chrysalis, orbiting between Titan and Iapetus in a 3:1 resonance with Titan, was torn apart roughly 100 to 200 million years ago, about 160 million years ago in the case they illustrate, freeing Saturn's tilt from Neptune's pull and making the rings of its debris. We draw that illustrated case, so the rings appear in the 66 Ma sky and after, and before them Chrysalis is a moon of magnitude 13 to 14 beside a ringless Saturn. Younger rings are drawn more massive and cleaner: their depth grows with youth as age$^{-0.75}$, between ring-rain losses (O'Donoghue et al. 2019; Waite et al. 2018) and viscous spreading (Salmon et al. 2010), half again today's at 66 Ma, and the meteoroid darkening of the C ring and Cassini division (Cuzzi & Estrada 1998) is two fifths of today's. Both are guesses. Saturn's pole precesses once in about 1.87 million years (Ward & Hamilton 2004), so the rings open from a different direction at each epoch: by a known angle at the Pleistocene supernovae, at random beyond ten million years.

6.5 Neptune's clouds

Neptune close up is drawn the same way, a tracer in its own flow. The winds are Voyager 2's cloud-tracked profile (Sromovsky et al. 1993): retrograde at 400 m/s at the equator, prograde at 250 m/s near 70°, in System III, the 16.11-hour radio period that also turns the planet. Its deck is smooth, so the eddies stir it only lightly and no billows roll up between bands. Its color is the true one, a pale greenish blue only a little bluer than Uranus's (Irwin et al. 2024), not the deep blue of Voyager's contrast-stretched images. On it are bright clouds of methane ice, in streaks drawn out by the shear of the winds that carry them, at southern mid-latitudes, near 30°N and near 70°S, each lasting some days; and the dark wave circling near 60°S, wandering a few degrees in latitude and drifting 4.87° a day in System III, with a bright band just north of it (Karkoschka 2011; Chavez et al. 2023). Dark vortices come and go, a new one every four to six years, each lasting about two (Hsu et al. 2019): deep-blue

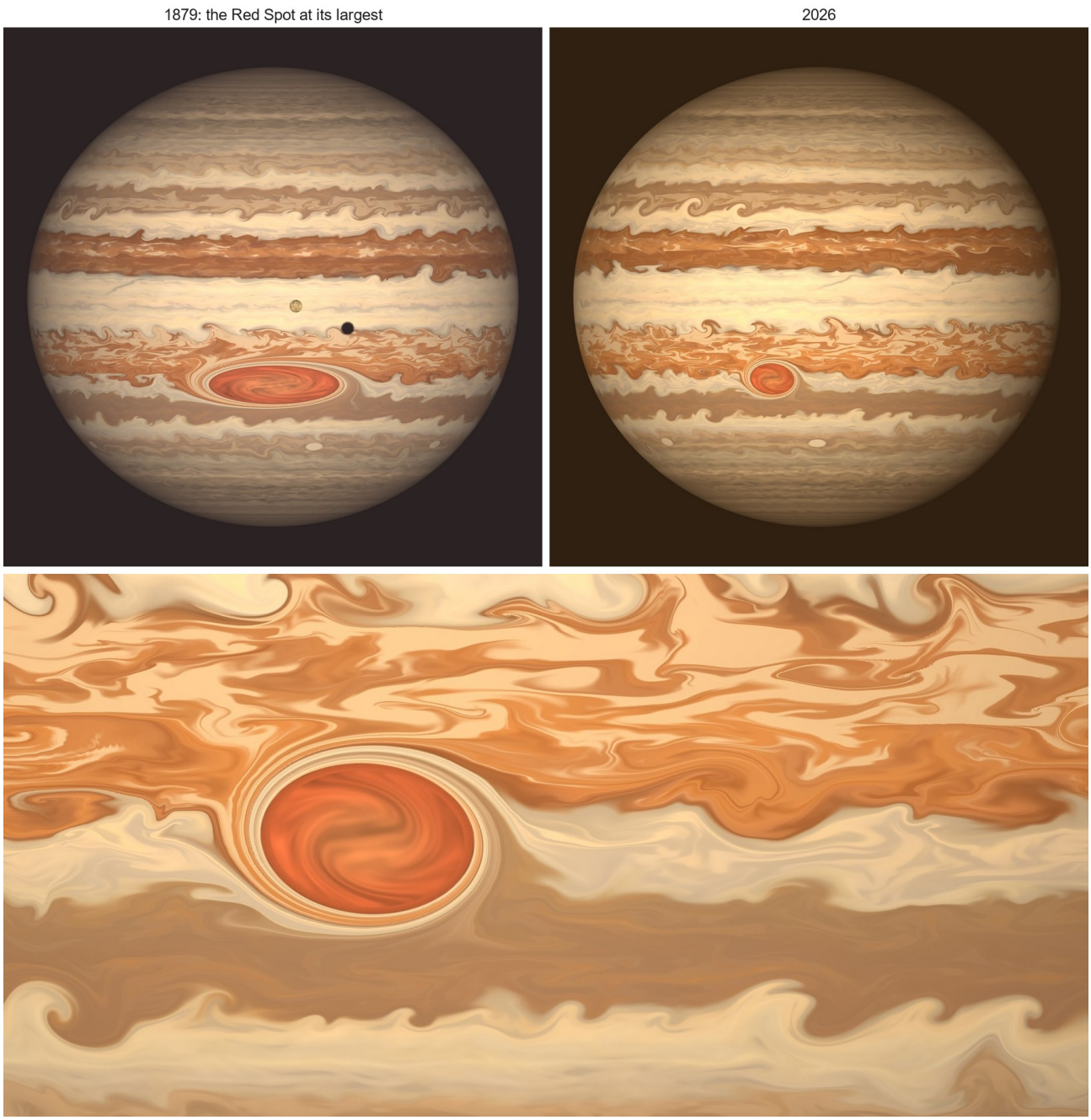


Figure 11: Jupiter close up, as the site's telescope view draws it, each view turned so the equator lies level, north up. Top: the Great Red Spot at its largest, about 35° long in 1879, when it was also reddest, and in 2026, about 12° long; in each Jupiter is shown on a night when the spot faces us. Bottom: the 2026 spot close up, a vortex in the same flow as the belts and zones around it, which eddies marble and Kelvin–Helmholtz billows roll along their edges.

ovals with bright methane clouds along their poleward edges, turning as anticyclones and drifting with the wind at their latitude. The Great Dark Spot that Voyager 2 found is drawn as it was, at 22°S, 13,000 by 6,600 km, from late 1988 until it was gone in 1994 (Smith et al. 1989). At other times a vortex is drawn at random as their rates and sizes have it, since where one was on any given day is not known.

6.6 The giants when young

The giant planets were warmer when young, and their clouds were different. Each cools as it ages: we take its intrinsic temperature, the effective temperature its own heat alone would give it, to fall as the age to the −0.29, a power-law fit to the cooling tracks of Burrows et al. (1997), scaled to today's (98 K for Jupiter, 78 K for Saturn, 53 K for Neptune, from the effective temperatures Voyager measured, Hanel et al. 1981, 1983 and Pearl & Conrath 1991, less the sunlight's share), and add the young Sun's light at each planet's distance. For planets of these masses the hot and cold starts of formation meet within a few tens of millions of years (Spiegel & Burrows 2012), so the history is surest over the last four billion years and least sure in the oldest sky. Then Jupiter's effective temperature was near 500 K in the 4.55 Ga sky, 330 K in the 4.5 Ga sky, 185 K at 4.0 Ga and 170 K at 3.8 Ga, against 124 K now; Saturn's 395, 265, 145 and 135 K, against 95; Neptune's 270, 180, 98 and 90 K, against 59. Uranus is drawn as it is now: it gives out too little heat for any model that starts it hot (Fortney et al. 2011), and how warm it was is not known.

A cloud deck forms where the air is cold enough for its gas to condense, and the coldest air, at the top of the troposphere, is near 0.85 of the effective temperature, so as each planet cooled its clouds formed in turn from the bottom up; young, Jupiter and Saturn had no ammonia clouds and a top deck of water clouds, and the ice giants' clouds lay higher (Fortney et al. 2011). We draw three stages. Above about 165 K Jupiter and Saturn have no ammonia ice on top and show the NH4SH haze beneath, paler and less colored, as Jupiter is at 4.0 and 3.8 Ga; above about 85 K Neptune has no methane ice, no bright streaks, and shows its H2S deck higher up, as pale a cyan as Uranus, whose top deck is H2S now (Irwin et al. 2018), as Neptune is at 4.0 and 3.8 Ga. Warmer still, above about 230 K for the gas giants and 155 K for the ice giants, the top is a high deck of NH4SH and water clouds, bright and near white, Sudarsky et al.'s (2000) class II: Jupiter and Saturn in the 4.5 Ga sky and the ice giants in both of the oldest, their bands showing more in brightness than in hue, the ice giants' storms of moist convection rising bright out of their mid-latitudes. Above about 380 K water no longer condenses high up and the planet is clear, their class III, a dark azure of Rayleigh scattering and methane absorption: Jupiter in the 4.55 Ga sky, and most of Saturn. Each planet's point is tinted and brightened or dimmed with its disk, and its spectrum's reflectance mixed toward the stage's: flatter and grey for the NH4SH and water decks, with shallower methane bands above the higher clouds, and blue and dark in the red, with deep methane bands, when clear.

Oceanus, of about Neptune's mass, cools as Neptune does; nearer the Sun, at 9.6 AU, its effective temperature is 270 K in the 4.55 Ga sky and 180 K in the 4.5 Ga sky, so in both it has the high bright deck of NH4SH and water clouds, a pale banded cyan. Its bands are laid out as the ice giants' circulation runs, air rising at mid-latitudes, where clouds are thick and bright, and sinking at the equator and poles, where they are thin (Fletcher et al. 2020): a broad darker equatorial belt, bright zones from about 18° to 45°, narrower belts near 50° and darker polar caps, carried and stirred as Neptune's are, in Neptune's winds, with bright storm tops in the rising zones of both hemispheres. Its day, unknown, is taken as 16 hours. Through the tens of bars of CO_2 of the 4.5 Ga sky none of this can be seen from the ground.

6.7 The planets' moons

The moons with fitted orbits, fitted here to JPL Horizons over 1980–2060, are Io, Europa, Ganymede and Callisto; Mimas, Enceladus, Tethys, Dione, Rhea, Titan and Iapetus; Miranda, Ariel, Umbriel, Titania and Oberon; Triton; and Mars's Phobos and Deimos. Most hold to a tenth of a degree. Iapetus's two faces show: its trailing side is ten times brighter than its leading side, so it is magnitude 10.2 at western elongation and 11.9 at eastern. The planets' moons also change with the epoch. Triton, retrograde, was captured in the giant-planet instability, so the two oldest skies lack it, and Neptune has instead a family of regular moons, here named Kymo, Galene, Thetis and Amphitrite, drawn like Uranus's four largest at the same distances: Rufu & Canup (2017) find that a system no heavier than Uranus's fits best, and that capturing Triton scattered and destroyed it. Mars, at 4.0 and 3.8 Ga, has Harmonia (named here), the moon of some twenty times Phobos's mass at 3.3 Mars radii that Ćuk et al. (2020) find tilted Deimos's orbit, one generation of Mars's cycle of rings and moons (Hesselbrock & Minton 2017). Phobos is falling toward Mars: its tidal acceleration, half the rate its mean motion grows, is 1.258×10^{-3} degrees per year squared (Brozović, Jacobson & Park 2025), which is put back into its fitted longitude, so its place stays right to a few hundredths of a degree across centuries rather than drifting by degrees. Back in time, with Mars's tidal lag held fixed, its orbit grows as the 2/13 power of 1 + t/38 Myr, to 3.2 Mars radii at 66 Ma against 2.77 now; Deimos, beyond the synchronous orbit, was only some 130 km nearer at 3.8 Ga. Both of Phobos's histories fit its orbit (Ćuk, Anand & Minton 2025): formed early just inside the synchronous orbit, or about 100 million years ago at the Roche limit as the latest moon of the ring cycle. The site follows the second, as its Harmonia does, so Phobos appears only in the skies of the last 100 million years, and Deimos from 4.3 Ga on.

7 The stars and the Galaxy

7.1 The stars moved back

Today's sky has the thousand brightest stars of the Yale Bright Star Catalogue (Hoffleit & Warren 1991), down to about magnitude 4.6, carried to the year shown by proper motion and precession, each tinted by its color temperature and shown only when it is brighter than the naked-eye limit for the sky around it, so stars come out one by one in twilight and cities keep only the brightest. Earth's axis keeps today's tilt and direction in every epoch, since its precession in the deep past cannot be pinned down. Going back, the stars move, and how far the page can follow them sets three kinds of sky.

For the Pleistocene supernovae, 1.78 Ma and 342 ka, every catalogue star is moved in a straight line from its Hipparcos parallax, proper motion and radial velocity (the XHIP compilation of Anderson & Francis 2012), and the thousand brightest at that time are kept. Over two million years the Galaxy's pull bends their paths by only a few parsecs. The sky is recognizable but rearranged. At 1.78 Ma Canopus is the brightest star, −1.6, and Sirius, far away then, is fourth magnitude; at 342 ka Aldebaran, near its closest pass, outshines everything at −1.5, and Sirius is +0.8. Constellation figures move with their stars, and a line is dropped once its stars have drifted three times today's distance apart.

At 66 Ma the Sun and the catalogue stars are traced back along orbits in a model Galaxy: a flat rotation curve of 233 km/s, a vertical oscillation of 84 million years, and the Sun's motion from Schönrich, Binney & Dehnen (2010), keeping only stars old enough to have been shining. Six of them were brighter than magnitude 4.6. Further back none of today's neighbors was near the Sun, and every star is a stand-in: a copy of a star of today's sky, keeping its brightness, color, spectral type and galactic latitude, at a random galactic longitude, a different random sky for each age. The stand-ins' labels give the type they copy, not a name. The sky's statistics are right; its patterns are invented, as they would be for any reconstruction that far back.

7.2 Stars brighter than Sirius

Sirius, at magnitude −1.46, is the brightest star in today's night sky, but it has not always been. The Sun's neighbours drift past it at tens

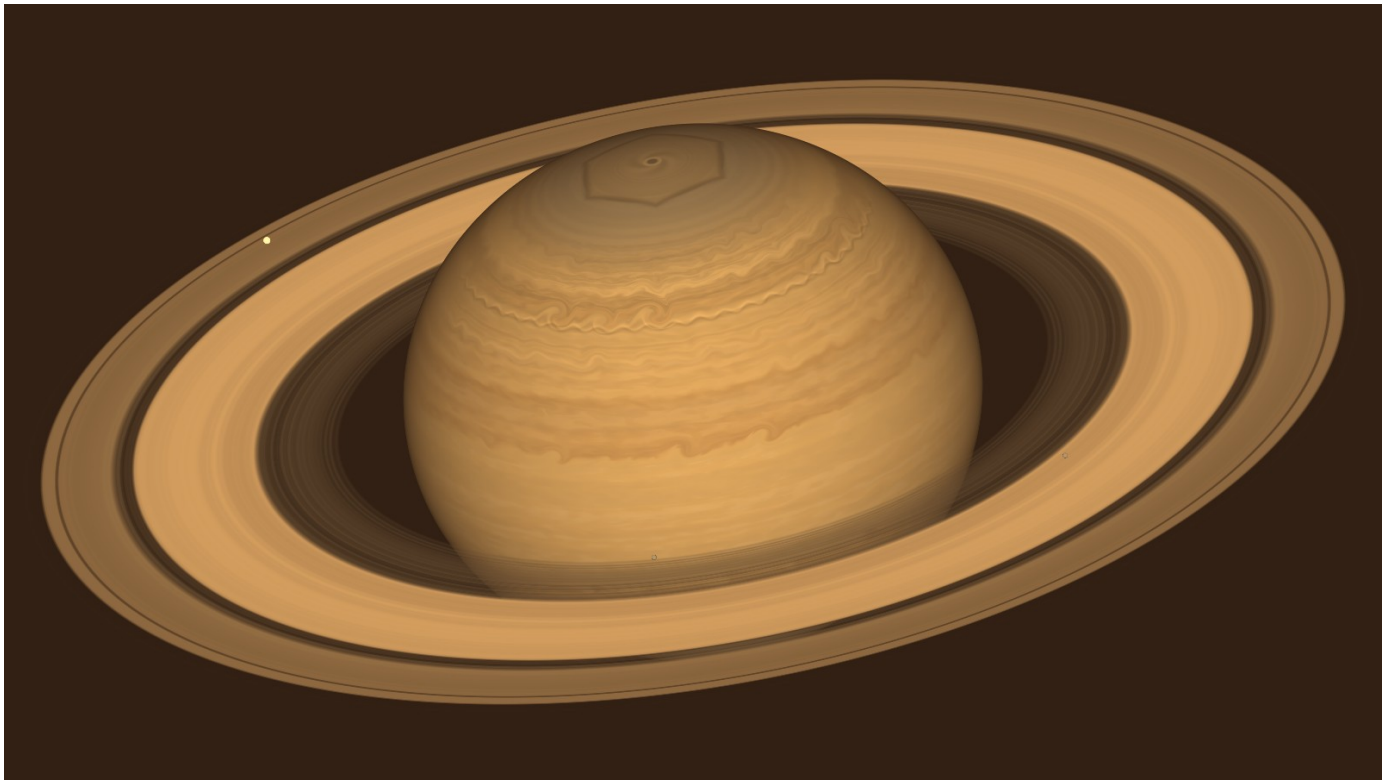


Figure 12: Saturn close up in the modern sky on 15 June 2017, at opposition, its rings filling the site's telescope view. The north pole is tilted 27° toward us, showing the hexagon around it, golden inside in the northern summer, with the polar vortex at its center. The belts are stirred by the flow drawn for Jupiter under Cassini's winds, calmer and muted by the haze. The rings are drawn from the Cassini occultation profile with single-scattering photometry; at opposition the globe's shadow on them lies hidden behind it.

of kilometres a second, so a star a hundred parsecs away now may have passed within a few parsecs of it a few million years ago. Tracing every Hipparcos star that has a radial velocity back 10 Myr through the Galaxy, with 2,000 draws over each star's measurement errors, finds eleven that outshone today's Sirius (Table 7). The brightest is θ Columbae, a fifth-magnitude star today, which passed about 2 pc from the Sun 4.7 million years ago and peaked near magnitude -5, as bright as Venus at its best; Gaia DR3 astrometry (Gaia Collaboration 2023) gives the same (-5.2 at 2.1 pc, 5.1 million years ago). ε Canis Majoris reached -4.35 4.4 million years ago (its spell as the brightest star has been known since Tomkin 1998, who gave -4.0 at 4.7 million years), and β Canis Majoris -3.6 at the same time. From about 3.5 to 5 million years ago the night sky held at least two stars brighter than any star today, and at times three.

These stars now lie in Columba, Canis Major and Puppis because that is the solar antapex: the Sun moves toward Hercules at about 20 km/s relative to its neighbours, so the stars it has passed end up behind it. The closest pass, ζ Leporis at 1.3 pc 850,000 years ago, came as close as α Centauri is now, far too distant to disturb the planets. Only distance changes the brightness here. ε and β Canis Majoris are giants 12 to 22 million years old and were probably a few tenths of a magnitude fainter then. Algol's radial velocity (3.7 ± 3.9 km/s, hard to measure in an eclipsing binary) is too uncertain to say whether it passed within 1.3 or 28 pc of the Sun. Rigel, Saiph, Mintaka, Alnitak and ξ Persei also come out bright, but 6 to 10 million years ago, about their own ages, so they are left out. The site's skies do not show these peaks: the supernova epochs (1.78 Ma and 342 ka) fall between them, though at 342 ka Aldebaran is close to its peak of -1.5.

7.3 The Milky Way, nebulae and galaxies

The Milky Way is a model in galactic coordinates rather than a photograph: a disk and bulge, the Sagittarius and Scutum star clouds, the Great Rift from Cygnus to Sagittarius, the Coalsack and the Ophiuchus and Taurus dust, peaking at 20.2 mag/arcsec^2 in the Sagittarius star cloud (Leinert et al. 1998), and shown only in a dark sky. In the older skies its brightest part lies where the Galactic center then lay as seen from the Sun, from the Sun's orbit traced back through the same model Galaxy, and the band shifts by a few degrees with the Sun's height above the plane: at 66 Ma the Sun was 100 pc north of it. Beyond about a billion years the Sun's place on its orbit is only nominal. The Galaxy's own structure is today's in every epoch.

The 34 nebulae and galaxies that binoculars show under a dark sky are drawn from survey photographs, their stars taken out, each scaled to its catalogue magnitude and surface brightness, and turning into its photograph when zoomed. They change with the epoch. The Andromeda and Triangulum galaxies fall toward the Milky Way on a timing-argument orbit (van der Marel et al. 2012; Kahn & Woltjer 1959), so at 4.5 Ga Andromeda is three quarters of today's size. The Large Magellanic Cloud is on its first infall (Kallivayalil et al. 2013), traced back through the Galaxy's halo with dynamical friction: a small patch in another part of the sky in the Hadean, larger than today's at 66 Ma. Galaxies beyond the Local Group were nearer by the cosmic scale factor. The nebulae are young: each is drawn only within its age, so the Orion Nebula is absent at 1.78 Ma, and none of today's shine at 66 Ma or before, when others, unknowable, did.

7.4 Supernovae

Two skies are lit by a supernova. Geminga's, about 342,000 years ago by its pulsar's spin-down age (Salvati & Sacco 2008), exploded 90 to 240 pc away toward Orion (Pellizza et al. 2005); a typical Type II-P peaks near absolute magnitude -16.75 (Richardson et al. 2014), so from 147 pc it reached about -11, a fifth of the full Moon's light from a single point, bright enough to show by day and cast shadows at night for the three months of its plateau. The ζ Ophiuchi supernova, 1.78 million years ago at 107 pc in the Scorpius–Centaurus association (Neuhäuser et al. 2020), unbound its companion, now the runaway star ζ Oph, and peaked near -11.6, twice as bright. Its spectrum readout is a Type II-P near peak, with P Cygni profiles from fast ejecta (Filippenko 1997). At these distances neither touches the sky's colors (Thomas et al. 2016). Betelgeuse is the same event still to come, in the same patch of sky: 1.4° from Geminga's star in the sky of 342 ka, a red supergiant of 16.5 to 19 solar masses burning helium in its core, some 100,000 years from collapse, at 168 pc (Joyce et al.

Table 7: Stars that outshone today's Sirius (V = −1.46) in the last 10 million years, traced back through the Galaxy from Hipparcos astrometry (XHIP; Anderson & Francis 2012) with 2,000 draws over the measurement errors. Peak magnitudes are medians with the 68% range; the last column is the fraction of draws brighter than Sirius today. Brightness changes with distance only.

Star	Type	Today: V, pc	Peak V (68% range)	Closest (pc)	Myr ago	Brighter than Sirius
θ Columbae	B8:IV	5.00, 221	-5.00 (-5.77 to -4.41)	2.2	4.73	100%
ε Canis Majoris (Adara)	B1III	1.50, 124	-4.35 (-4.44 to -4.26)	8.4	4.42	100%
β Canis Majoris (Mirzam)	B1III	1.98, 151	-3.58 (-3.72 to -3.45)	11.7	4.35	100%
ρ Orionis	K0 III	4.46, 105	-3.16 (-4.24 to -2.40)	3.1	2.54	100%
Algol	B7V	2.09, 28	-2.88 (-4.65 to 2.09)	2.9	5.07	62%
ζ Leporis	A2IV-V(n)	3.55, 22	-2.54 (-2.63 to -2.43)	1.3	0.85	100%
ν Puppis	B7III	3.17, 114	-2.42 (-2.64 to -2.22)	8.7	3.56	100%
ζ Sagittarii (Ascella)	A2.5Va	2.60, 27	-1.97 (-2.07 to -1.87)	3.3	1.05	100%
Canopus	A9II	-0.62, 95	-1.92 (-2.01 to -1.82)	52.2	3.18	100%
Aldebaran	K5III	0.87, 20	-1.50 (-1.53 to -1.47)	6.8	0.33	92%
ζ Canis Majoris (Furud)	B2.5 V	3.02, 111	-1.47 (-1.87 to -0.94)	14.0	3.27	51%



Figure 13: The Geminga supernova 342,000 years ago, near magnitude −11 in Orion, a fifth of the full Moon's light from a single point, among the stars moved back along their measured motions.

2020), or 222 pc by radio astrometry (Harper et al. 2017). A red supergiant ends as a Type II-P, so it should peak at −10 to −10.6, close to Geminga's −11, and the Geminga sky is a fair picture of Orion's own to come.

8 The night and what crosses it

8.1 Twilight and the night sky

Because the photometric scale of Section 2 is absolute, the night is drawn on the same scale as the day. The sky is computed for Sun heights down to 20° below the horizon; past 9° below, where the model's twilight fades too slowly, we make it dim by about a magnitude per degree, as measured. Without the Sun or Moon the sky has its own light: today 22.0 mag/arcsec^2 at a dark zenith (Leinert et al. 1998), half of it starlight and zodiacal light and half airglow, brighter toward the horizon.

Nearly all of today's airglow needs free oxygen: the 557.7 nm green and 630 nm red lines of oxygen atoms, the hydroxyl bands (H + O_3), the sodium D lines cycled through ozone, and the NO_2 continuum. We take it as 90% of today's at 466 Ma, 80% in the Snowball epoch (169 DU of ozone and less O_2), and 30% at 2.2 Ga, where 1% of today's oxygen leaves few atoms near 95 km. Before the Great Oxidation it is gone, replaced by a faint violet glow of the O_2 Herzberg II bands from oxygen split off CO_2, as on Venus (Krasnopolsky 1983): about as many photons as today's airglow but only 15% of its luminance, so Archean and Hadean nights were darker and bluer. These strengths are estimates; no model of an anoxic nightglow exists.

Starlight, the Milky Way and the airglow all come from above the air and are dimmed by it, at the rate given in Section 2: 0.25 magnitude per airmass in clean air, 0.8 under the thick haze or a polluted city, 0.85 under the proto-Earth's guessed air and dust, 2.3 through the impact-winter soot, and 5.8 under 30 bar of CO_2. That alone would leave the brightest stars at fourth magnitude overhead, but the bombardment's zodiacal glow, two thousand times today's, hides every one.

8.2 Aurorae

The aurora is an oval round the magnetic pole, fixed relative to the Sun: on a moderately active night today (Kp 3) it spans about 63–72° latitude on the night side (the ovals of Feldstein & Starkov 1967). Its field lines reach out to the magnetopause, whose distance goes as the sixth root of the field's pressure over the solar wind's. At 3.45 Ga a field half to 70% of today's and the young Sun's dense wind held the magnetopause at about half today's distance (Tarduno et al. 2010), which brings the oval's edge down to about 50° even in quiet times, and the young Sun's frequent flares and mass ejections (Airapetian et al. 2016) make storms the usual state. Through the Proterozoic the wind weakens (Wood et al. 2005) and the oval retreats toward the pole.

The colors follow the air. Today's green and red lines are oxygen atoms. At 2.2 Ga they are about a third of today's. With no free oxygen the aurora is nitrogen's: violet N_2^+ bands and a pink border of N_2 first positive bands where faster electrons reach lowest, brighter without oxygen to quench them, with a weak green line from CO_2 split by the electrons, as Perseverance saw on Mars (Knutsen et al. 2025). The 30-bar Hadean air is mostly CO_2, so there the green line would be stronger and the nitrogen faint, but the air, at 5.8 magnitudes of extinction per airmass, and the bombardment's zodiacal glow hide the aurora altogether. Brightness is for a dark-adapted eye, which sees violet far better than the deep red. The ancient line strengths are estimates.

Each night has its own activity about the epoch's typical level, and the oval is drawn at it. Today that is Kp 3; the young Sun's typical night is far stormier, Kp 6 at 4.0 Ga, and its oval reaches correspondingly lower, to about 37° on the night side at 4.0 Ga and 43° at 3.8 Ga (Table 8 gives both). The arcs are thin sheets along the field lines, green below and red above, with field-aligned rays converging overhead on the magnetic zenith; they fold, drift and break up in two substorms a night near midnight. A storm of the class of September 1859, when aurora stood overhead near 40° magnetic latitude, brings today's oval down to about 45° and the young Earth's to the equator.

8.3 Meteors and meteor showers

Each meteor is a meteoroid entering the air along a straight path 80 to 125 km up, with its brightness, size and speed tied by the photographic relation of Jacchia, Verniani & Briggs (1967) and its count set by the measured fireball rate. Today a dark site sees about 8 an hour, more before dawn. A meteor shows only when brighter than the sky around it, so twilight, moonlight and city light thin them out. The air enters three ways. Bright, fast meteors leave persistent trains that glow green and then orange; that chemistry needs oxygen and ozone, so there are no trains before the Great Oxidation. Each meteor's color comes from a spectrum built for its speed and the epoch's air: the metals of the meteoroid, a hotter component that grows with speed, and the air's own emission, oxygen and nitrogen today or carbon monoxide bands in a CO_2-rich air, which no one has measured. And in the Early Hadean's 30 bar every density level sits higher, so meteors burn about 22 km higher up.



Figure 14: An aurora over the young Earth at 3.8 Ga, from mid-latitudes, on an active night, drawn as a dark-adapted eye sees it: with no free oxygen, nitrogen's violet bands and pink lower border rather than today's green and red. The aurora is brightened for print, 100 times its luminance on the site, where a true night is too dim to reproduce on paper.

The rate follows the lunar cratering record (Neukum et al. 2001): about 10,000 times today's at 4.5 Ga, extrapolated past the record's oldest ages, in the giant impact's debris, 500 at 4.0 Ga and 120 at 3.8 Ga, of slow, yellow-orange fireballs from a main-belt-like population (Strom et al. 2015; Marchi et al. 2014), with impact flashes on the closer Moon's dark side; twice today's at 2.7 Ga; and a hundred times today's fireballs at 466 Ma, after the L-chondrite parent body broke up (Schmitz et al. 2019).

Meteor showers come on their own dates. The modern-era skies have the 38 showers of the IMO's working list, apart from its Antihelion source (Rendtel 2027, Table 5), with their radiants and drift, speeds, population indices and peak rates, rising and falling about the maximum as Jenniskens (1994) measured them, and their outbursts and storms at the hour they were seen: the Leonids of 1799 to 2002, the Andromedids of 1872 and 1885, the Draconids of 1933 and 1946 and others (Jenniskens 1995 and the IMO's analyses), with the Leonid peaks predicted for 2034. Showers last thousands of years, so the older epochs have their own, drawn at random with the geometry, speeds and profiles of today's and outbursts at today's rate, about 0.5 a year above ZHR 100 and one in 45 years above 10,000. We take their number to follow the comets that feed them: three quarters from short-period comets, a quarter from long-period ones (only about a fifth of known streams have an identified parent; Ye & Jenniskens 2024), giving about 30 times today's at 4.5 Ga, 5 at 4.0 Ga, today's by 2 Ga and 1.75 at 1.78 Ma, in the comet shower after a star's passage. No published work estimates showers in the early Solar System, so these are estimates, and in the bombardment they hardly show against the sporadic rate.

8.4 Fireballs and impacts

The largest meteoroids do not burn up. Bodies a meter across and more come in as superbolides, brighter than magnitude -15, and leave dust trails from about 80 km down to where they break up at 25 to 40 km, which drift and shear in the high winds, as Chelyabinsk's did for over an hour (Borovička et al. 2013); today about one comes within 1,100 km of a given place every three years, and in the bombardment several an hour. Bodies brighter than about -24 reach the ground. When one strikes within 200 km it is drawn with its fireball, a thermal pulse cooling from 6,500 to 2,000 K, the blast wave, a condensation ring and the rising cloud, scaled from nuclear-test and impact-effect relations (Glasstone & Dolan 1977; Collins, Melosh & Marcus 2005). Today that happens to a given place about once in 50,000 years; in the bombardment, constantly. In the bombardment epochs, impact flashes also show on the closer Moon's night side, at today's measured rate scaled up (Suggs et al. 2014).

8.5 Dust, comets and the Ordovician ring

The zodiacal light, sunlight off interplanetary dust, is part of today's natural night sky. Where the inner Solar System held more dust the excess is drawn along the ecliptic, with the gegenschein opposite the Sun: 30 times today's after the L-chondrite break-up and 50 to 2,000 times during the bombardment (estimates after Nesvorný et al. 2010), as bright as a moonlit sky, hiding the fainter stars and meteors. At 466 Ma the speculative debris ring of Tomkins, Martin & Cawood (2024) is drawn between 1.55 and 2.75 Earth radii with a normal optical depth of at most 0.00015: lost in the blue by day, a low arch at night cut by the Earth's shadow, and at the equator a source of slow, grazing fireballs from the west.

Comets in the modern-era skies are the real ones on their real dates, from JPL's orbits; elsewhere they are drawn at the rate of their time, about one naked-eye comet a year today, 30 times that at 4.5 Ga while the giant planets eroded and then scattered the comet disk, falling to today's level by about 2 Ga, and about half of today's in the quiet Phanerozoic epochs, because the present flux of new long-period comets is still about two to two and a half times the tide-only rate (Kaib & Raymond 2026). Their heads and tails are dimmed by each epoch's air like the stars.

9 The sky people made

9.1 City light

In the modern epochs the observer stands in a mid-sized city: a zenith glow of 18.8 mag/arcsec^2 today (Falchi et al. 2016), with orange sodium light in 1980–2000, at the brightness of the first World Atlas (Cinzano, Falchi & Elvidge 2001), 1.6 times today's in polluted air, held at today's level in 2100 (the recent 9.6% a year, Kyba et al. 2023, cannot be projected that far), and almost nothing from the oil lamps of 1815 (Fouquet & Pearson 2006). The glow rises about threefold toward the horizon, and cloud bases over the city glow orange several times brighter than the clear sky (Kyba et al. 2011). Against that glow the city keeps only the brightest stars, and none of the Milky Way.

9.2 The sky before electric light

The 1815 sky is the last before gas and electric street lighting. Its towns burn oil lamps and candles: Britain then used about one six-thousand-five-hundredth of the artificial light per head that it used in 2000 (Fouquet & Pearson 2006), so the town's glow on the sky is some ten thousand times fainter than a modern city's, and the Milky Way stands over the rooftops.

9.3 Aircraft

The modern-era skies have airliners and their contrails: about 1.1 aircraft over each 10,000 km^2 of busy mid-latitude sky by day, from European traffic statistics, at 9 to 12 km along a few airways, quieter after midnight; under half of today's traffic in 1990, twice today's in 2100, none in 1815. A contrail forms where the exhaust's mixing line crosses saturation (Schmidt 1941; Appleman 1953), most often at mid and high latitudes, and lives seconds in dry air but spreads into contrail cirrus lasting hours in the 10 to 20% of cruise levels that are supersaturated with respect to ice (Gierens et al. 1999; Schumann 2005; Minnis et al. 1998). At night the aircraft show as their navigation lights, beacon and strobes.

9.4 Satellites

The 2100 sky has today's air and the satellites that may fill it. Lawler, Boley & Rein (2022) put every filed megaconstellation on orbit, 65,262 satellites in 30 shells from 328 to 1,200 km, and found the worst naked-eye light pollution near 50° latitude; each is drawn as their diffuse sphere of 0.8 m^2 effective area, so its magnitude is on the stars' scale. At the equinox a mid-latitude sky has a few hundred of them above naked-eye brightness in the hour after sunset, and the Earth's shadow takes them by midnight. On top of that fleet are the orbital datacenters modeled by Boley, Lawler & Rein (2026). Of their three filed designs



Figure 15: The proto-Earth's bombardment, at 4.55 Ga, from mid-latitudes, on a moonless world. Left: the whole sky at one instant of a midnight on the site's fisheye dome (zenith in the middle, horizon at the rim, north up), with about 18 meteors in the air at once, at 30,000 times today's rate, most of them slow asteroidal fireballs. Right: a ground strike in the first-person view, 6.8 seconds into an 8.5-second fall at 13.5 km/s: a stony body about 120 m across, magnitude −26.4, about as bright as the Sun, coming down 110 km away with 70 megatons, its dust trail behind it, and its light filling the night sky with the Sun 22° below the horizon. The weather's clouds are taken away for this view; the dust trail is drawn by the same cloud pass.

Table 8: Other phenomena by epoch. Refraction is the horizon bending relative to today's, from the epoch's gases. Airglow and aurora strengths are estimates; the aurora's latitude is the night-side equatorward edge of the oval, from the magnetopause distance, at Kp 3 and at the epoch's typical activity. Meteor and zodiacal-light rates are relative to today's.

Epoch	Refraction	Airglow	Starlight extinction (mag/airmass)	Aurora	Meteors	Zodiacal light
4.55 Ga proto-Earth	5.7× (trapped)	violet CO_2 glow only	0.85	from 35° (Kp 3), the equator (Kp 7); green CO_2 line, faint N_2	30,000×	10×
4.5 Ga Hadean	47× (trapped)	violet CO_2 glow only	5.8; no stars in the zodiacal glow	hidden by the air	10,000×	2,000×
4.0 Ga Hadean	1.8×	violet CO_2 glow only	0.35	from 48° (Kp 3), 37° (Kp 6); violet and pink N_2	500×	300×
3.8 Ga Archean	0.97×	violet CO_2 glow only	0.25	from 51° (Kp 3), 43° (Kp 5.5); violet and pink N_2	120×	50×
2.7 Ga Archean	0.90×	violet CO_2 glow only	0.25–1.8 with the haze	from 55° (Kp 3), 49° (Kp 5); violet and pink N_2	2×	1.5×
2.2 Ga post-GOE	0.85×	oxygen lines at 30%	0.25	from 60° (Kp 3), 58° (Kp 4); O lines at a third	1.3×	today
700 Ma Snowball	0.83×	oxygen lines at 80%	0.25	from 63° (Kp 3), 62° (Kp 3.3); O lines at 80%	0.6×	today
466 Ma Ordovician	0.97×	oxygen lines at 90%	0.25	today's	100× fireballs	30×
300 Ma Carboniferous	1.11×	today's	0.25	today's	0.5×	today
66 Ma impact winter	1.0×	today's, dimmed by soot	2.3	today's	today	today
1815 volcanic year	1.0×	today's	0.7	today's	today	today
Today	1.0× (34′ at the horizon)	22.0 mag/arcsec^2 natural sky	0.25	63–72° at Kp 3; O green and red	8 an hour	—

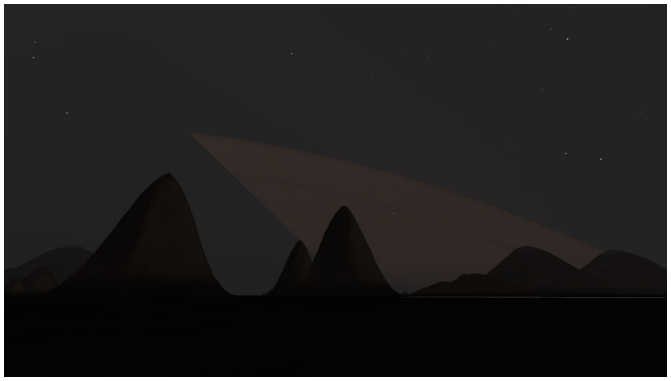


Figure 16: The night at 466 Ma, after the L-chondrite parent body's break-up: a hundred times today's fireballs, a bright zodiacal band, and the speculative ring of Tomkins, Martin & Cawood (2024) as a faint arch.

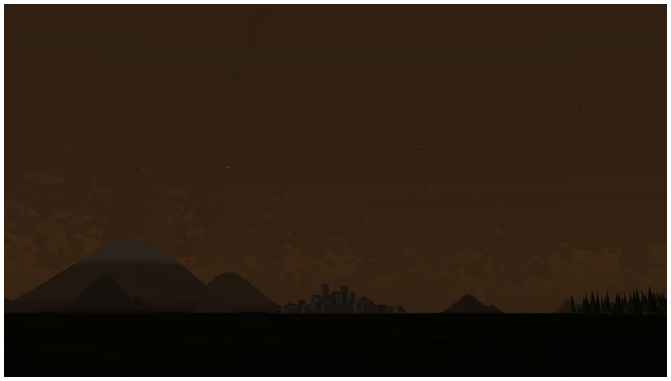


Figure 17: A city night today, looking toward the city 2.8 km away, its towers along the horizon: sodium and LED glow at 18.8 mag/arcsec^2 hides all but the brightest stars.



Figure 18: A town night in 1815: oil lamps light the streets but barely touch the sky, and the Milky Way shows over the hills.



Figure 19: Dusk in 2100: the filed megaconstellations and the Sunrise orbital datacenters, still in sunlight above the darkening sky.

we take Blue Origin's Sunrise and the relaxed one of their two node spreads: 51,600 Sun-synchronous satellites from 500 to 1,800 km in a dawn–dusk cross of rings whose nodes wander 10° to either side of the terminator. Each is a diffuse sphere of 800 m^2 at albedo 0.2, the size they call conservative next to panels several times larger: 160 m^2 of effective area, two hundred times a communications satellite's. An hour after sunset, thousands of them are above naked-eye brightness, a bright band along the terminator, each crossing the dome in minutes. The paper's all-night winter rings belong to the December solstice, which the page's equinox geometry does not show.

10 Epoch by epoch

The panels that follow are laid out identically so they can be compared at a glance. Each begins with the epoch's name and a one-line summary of the atmosphere assumed (Table 2 gives the full parameters). The upper row shows the noon sky dome at three latitudes, the equator, a mid-latitude site and the summer pole, with the zenith at the top of each swatch and the horizon at the bottom, the correlated color temperatures of both, and the Sun's disk drawn in its own color at its noon elevation, with a brightness that reflects how much of it survives the atmosphere. The two strips beneath trace the sky from the solar horizon across the zenith to the antisolar horizon with the Sun on the horizon and 4° below it. All swatches are shown at a brightness relative to today's clean sky, so a dim epoch reads as dim. The paragraph after each panel describes that epoch's sky and what else was in it.

Proto-Earth, ∼4.55 Ga

before the Moon-forming impact; a guessed ∼3 bar CO_2/N_2 air with impact dust; the young Sun at 17 Myr

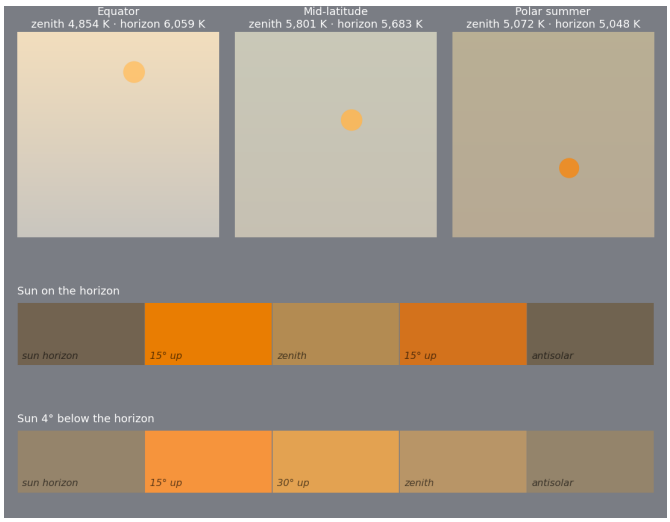


Figure 20: Proto-Earth, ∼4.55 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Some 17 million years after the first solids, the gas of the Sun's disk has just cleared, and with it, in the reading of the meteorites' bombardment ages (Edwards et al. 2024; Liu, Raymond & Jacobson 2022), the giant planets went unstable at about this time, some 5 to 20 million years after the first solids; we draw them just before it. Jupiter and Saturn sit in 3:2 resonance near 5.6 and 7.3 AU, and beyond them three ice giants: Uranus and Neptune, nearer than now, and between them and Saturn a fifth giant like Neptune, named here Oceanus, which Jupiter will throw out of the Solar System (Nesvorný 2011; Nesvorný & Morbidelli 2012). Neptune still has its own family of regular moons, named here Kymo, Galene, Thetis and Amphitrite; capturing Triton will scatter and destroy them (Rufu & Canup 2017). There is no Moon: the Earth is still growing, and Theia, the Mars-sized body that will strike it and make the Moon, is a wandering planet near the Earth's orbit, brighter than Venus when it passes close. The Sun is a pre-main-sequence star 17 million years old (Baraffe et al. 2015): 4,600 K, half today's light, a cooler, oranger disk 7% larger. Nothing constrains the proto-Earth's air; we take it as 3 bar of CO_2 over a bar of nitrogen, with dust from the impacts, so the sky can be seen through it. Where each planet was on any day cannot be known.

Early Hadean, ∼4.50 Ga

∼10 Myr after the Moon-forming impact: magma ocean frozen, ∼30 bar CO_2 left; Moon at ∼8 Earth radii

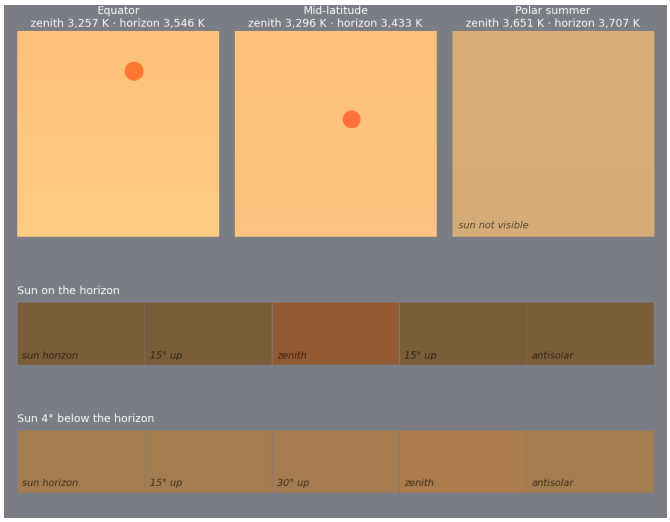


Figure 21: Early Hadean, ∼4.50 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

About ten million years after the Moon-forming impact. The magma ocean has frozen, the steam of the impact has rained out into the first ocean, and tens of bars of CO_2 are left in the air, which we take as 30 bar. A ∼30-bar CO_2 atmosphere has a Rayleigh optical depth near 5 at 550 nm and about 18 in the violet, so blue light is scattered back to space many times before it reaches the ground. What arrives is a diffuse, shadowless, peach-white glow of ∼3,300 K, nearly identical from zenith to horizon, under a young Sun that is only 68% as bright. The Sun itself is a dim ember at the equator and vanishes entirely at mid and high latitudes: this is a Venus-like sky. Sunset is a brief, deep-orange dimming of the whole dome rather than an event at the horizon. Above it, the Moon is still only about 8 Earth radii away (Korenaga 2023): 4° across, seven and a half times its width today and 56 times its area, its own magma ocean still open, glowing orange through cracks in its crust, and going round the Earth in 32 hours in the plane of the equator, while the Earth turns in under 6. It crosses the sky in hours, and for weeks around each equinox it eclipses the Sun at every new Moon, somewhere on the Earth, and passes through the Earth's shadow, 14° across at its distance, at every full Moon. Tidal recession will carry it out to about 39 Earth radii by 4.4 Ga. For this epoch we assume the giant planets' instability came late enough, near 4.5 Ga, that the fifth giant, Oceanus, and Neptune's first moons are still here; the bombardment ages favour an earlier one (Edwards et al. 2024), as in the 4.55 Ga sky. Through 30 bar of CO_2 and the glow of the impact's debris no star shows; the planets and their moons are there to be found by zooming in. The Sun is 67 million years old, near the main sequence: 5,500 K and 68% of today's light.

Late Hadean, ∼4.0 Ga

CO_2 mostly drawn down into carbonate; ∼1 bar N_2, ∼0.5 bar CO_2

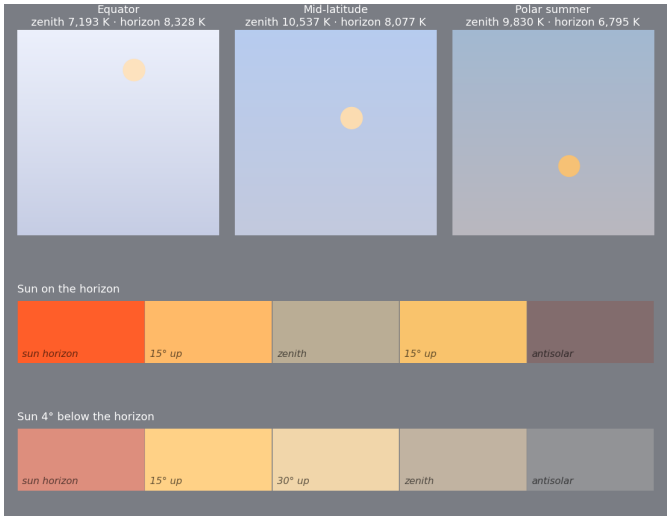


Figure 22: Late Hadean, ∼4.0 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Once most of the CO_2 has gone into carbonate, Rayleigh scattering falls to roughly twice the modern value. The sky turns pale blue at the zenith but clearly whiter than today (7,000–10,500 K versus 9,000–19,000 K), because the extra scattering fills the dome with multiply-scattered white light. Horizons are milky. The setting Sun is deep red and the twilight zenith, with no ozone yet, is cream rather than blue.

Early Archean, ∼3.8 Ga

haze-free, anoxic; CO_2 ∼0.1 bar, CH_4 ∼1000 ppm, no ozone

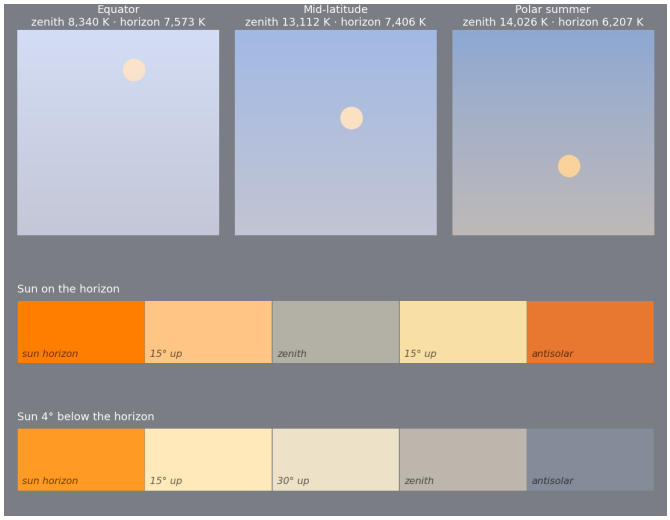


Figure 23: Early Archean, ∼3.8 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

With CO_2 down to ∼0.1 bar, noon skies are close to modern blue, only slightly whiter. The difference is at dusk. The blue of the modern twilight zenith is produced by ozone absorbing yellow-orange light (the Chappuis band) along the long grazing path; with no free O_2 there is no ozone, so the Archean twilight zenith is a pale yellow-cream, and the whole sky fades from blue to beige to orange as the Sun sets. Sunsets themselves are much like today's, slightly redder from the extra CO_2 and dimmer Sun.

Neoarchean, ∼2.7 Ga (thin haze)

biogenic CH_4/CO_2 ∼0.06: intermittent thin organic haze

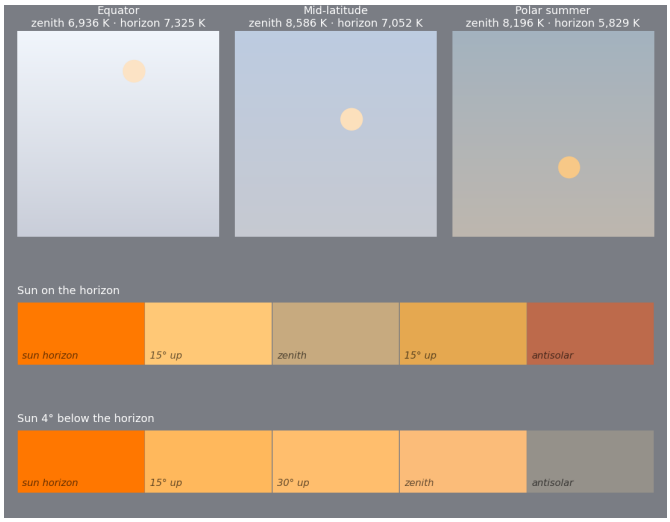


Figure 24: Neoarchean, ∼2.7 Ga (thin haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

An intermittent organic haze at optical depth ∼0.15 pushes the zenith toward a pastel, milk-blue (7,000–8,500 K) and lays a warm cast on the horizon; at the poles, where sunlight crosses the haze obliquely, the horizon turns visibly cream. Twilight is orange all the way to the zenith.

Neoarchean, ∼2.7 Ga (thick haze)

"Pale Orange Dot": Titan-like fractal organic haze, τ_{550}∼0.6

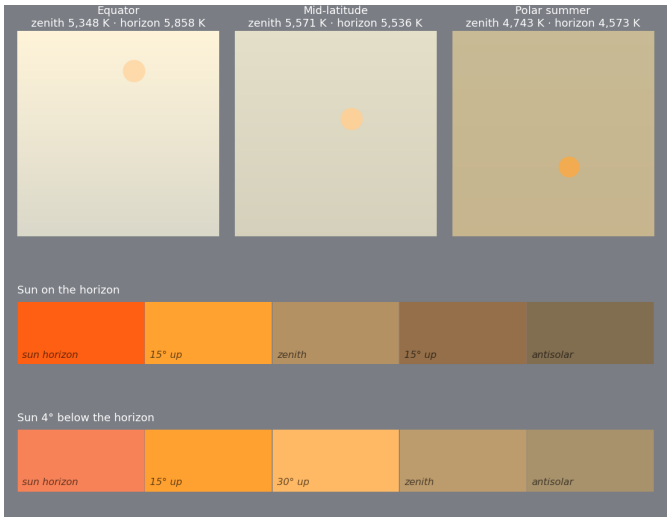


Figure 25: Neoarchean, ∼2.7 Ga (thick haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

This is the "Pale Orange Dot" seen from below. At optical depth ∼0.6 the haze absorbs blue and scatters forward, so the entire dome is a warm cream to pale tan (∼5,300 K at the equator, ∼4,700 K at the poles) with almost no zenith-to-horizon gradient: the sky looks like a bright overcast made of orange-tinted glass. The Sun is a soft, defined disk (haze scatters strongly forward) at 4,100 K. Sunsets are deep and long: the solar horizon reaches vermilion, the sky 15° above the Sun is amber, and the antisolar sky is peach rather than pink.

Neoarchean, ∼2.7 Ga (very thick haze)

upper-end haze, τ_{550}∼1.5: CH_4/CO_2 ∼0.2

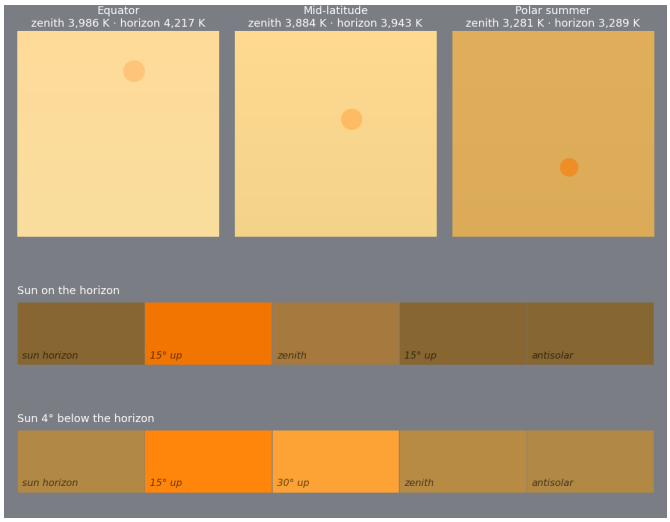


Figure 26: Neoarchean, ∼2.7 Ga (very thick haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

At the upper end of plausible haze ($\tau \approx 1.5$) the sky is a uniform saturated apricot (3,300–4,200 K), as on Titan, and the polar Sun is reduced to an orange smear. The ground under it gets roughly 20–30% of a clear sky's sunlight.

Paleoproterozoic, ∼2.2 Ga

after the Great Oxidation Event: O_2 ∼1% PAL, ozone column 66 DU

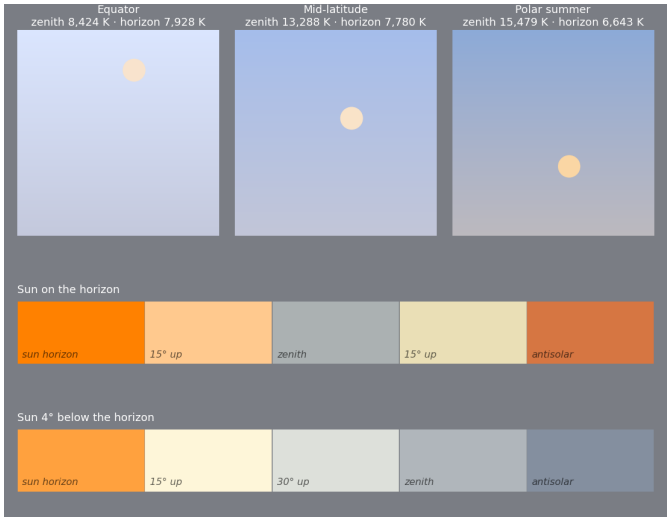


Figure 27: Paleoproterozoic, ∼2.2 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

After the Great Oxidation Event, about 1% of today's O_2 builds an ozone column of 66 DU, the Cooke et al. (2022) global mean for that oxygen level: about a fifth of a modern mid-latitude column, not the half that older estimates gave. Noon skies are already modern blue. Twilight is where the thin column shows. With the Sun 4° below the horizon the zenith leaves the cream of an ozone-free dusk (∼6,000 K) and becomes a pale blue-gray (∼7,100 K). It is not yet the deep blue of a 300 DU twilight (∼12,500 K). That saturated blue dusk is in place once the column is near modern, as in the snowball sky at 169 DU. (The pink antisolar arch needs no ozone: it is a Rayleigh effect and shows up faintly even in the Hadean model.)

Snowball Earth, ∼700 Ma

cold, dry, very clean air; ice/snow surface (albedo 0.75); ozone 169 DU

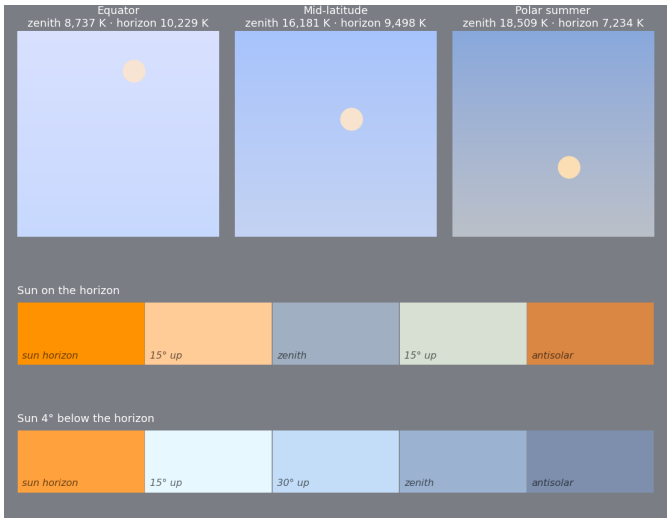


Figure 28: Snowball Earth, ∼700 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A frozen planet has dry, clean air and a surface albedo near 0.75, and together they give the bluest mid-latitude sky in Earth's history: 9,000 K at the equatorial zenith, 16,000 K at mid-latitudes and 18,500 K over polar ice, with horizons that stay distinctly blue instead of whitening, because the sunlit ice floods the sky from below with blue-white light. Sunsets are golden and comparatively pale; there is little aerosol to redden them.

Ordovician meteor storm, ∼466 Ma

after the L-chondrite parent body broke up: O_2 ∼17%, CO_2 ∼2,500 ppm, a little meteoric dust aloft

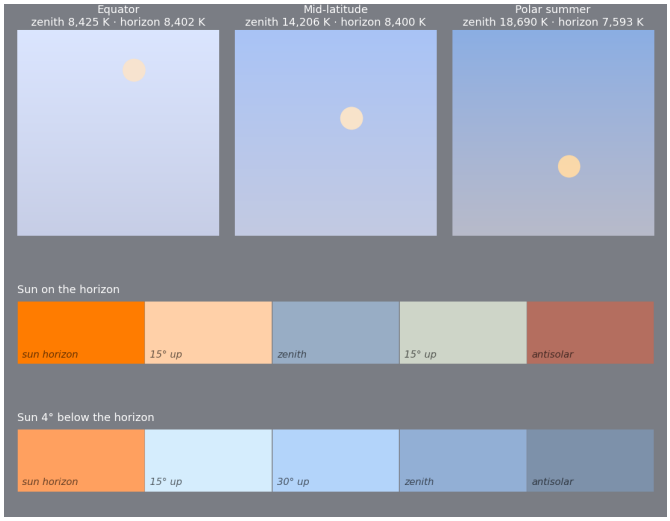


Figure 29: Ordovician meteor storm, ∼466 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

About 466 million years ago the L-chondrite parent body, an asteroid about 150 km across, was shattered in the main belt, and its fragments still make up almost a third of the meteorites that fall today. For more than two million years the fine dust reaching Earth rose a thousandfold or more, and decimetre stones about a hundredfold: fossil meteorites lie scattered through Swedish limestones of that age, and the dust may have helped tip the climate toward the mid-Ordovician ice age (Schmitz et al. 2019). The meteors here come a hundred times as often as today, most of them slow and yellow with sodium, and we take the zodiacal light as thirty times today's, a glowing band along the ecliptic. The ring is speculation: all 21 known Ordovician craters lie within 30° of the palaeo-equator, which led Tomkins, Martin & Cawood (2024) to propose that a large fragment broke up inside the Roche limit into a ring that lasted tens of millions of years; it is drawn as a faint, sunlit band that Earth's shadow cuts at night, with its debris falling at the equator as slow fireballs from the west. The air itself is close to today's: about 17% oxygen, 2,500 ppm CO_2, an ozone layer within a few percent of the modern one, under a Sun 4% fainter, so the day sky is today's blue.

Carboniferous, ∼300 Ma

O_2 ∼30-35%; total pressure ∼1.1 bar; wildfire smoke common

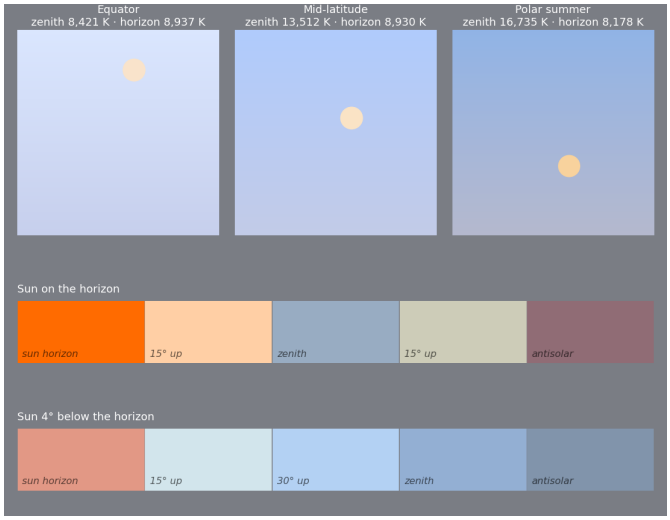


Figure 30: Carboniferous, ∼300 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Oxygen at 30–35% raises the total pressure and Rayleigh scattering by ∼10%, so skies are marginally brighter and horizons a touch whiter than today. Frequent wildfire smoke in the high-O_2 world would have produced red suns and brown-orange horizons far more often than in the modern era.

K-Pg impact winter, 66 Ma

months after Chicxulub: stratospheric soot (tau∼1.5) + sulfate (tau∼0.5)

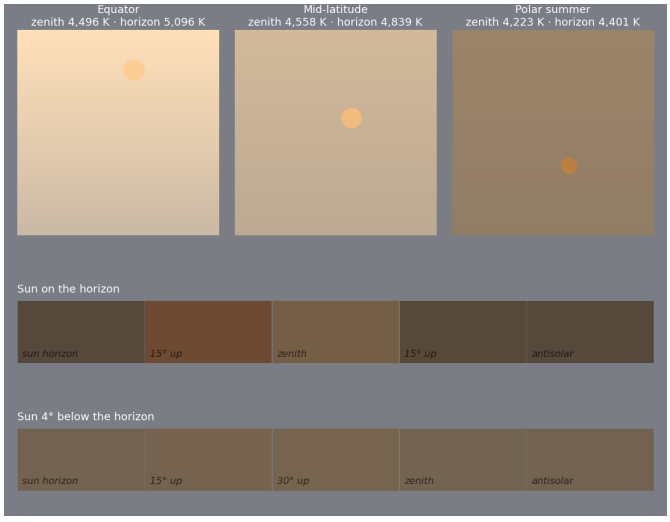


Figure 31: K-Pg impact winter, 66 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Months after the Chicxulub impact, a global stratospheric layer of soot and sulfate turns the sky into a dim, uniform amber-beige (4,200–5,100 K) with no blue at all, about one-fifth as bright as a clear sky at the equator and 20× dimmer at high latitudes. The Sun is a pale orange disk at low latitudes and effectively gone near the poles. Sunsets do not happen as color events: the whole dome simply fades to dark brown, and the twilight sky is gray.

ζ Ophiuchi supernova, ∼1.78 Ma

clean Early Pleistocene air, under a magnitude −11.6 supernova in Scorpius

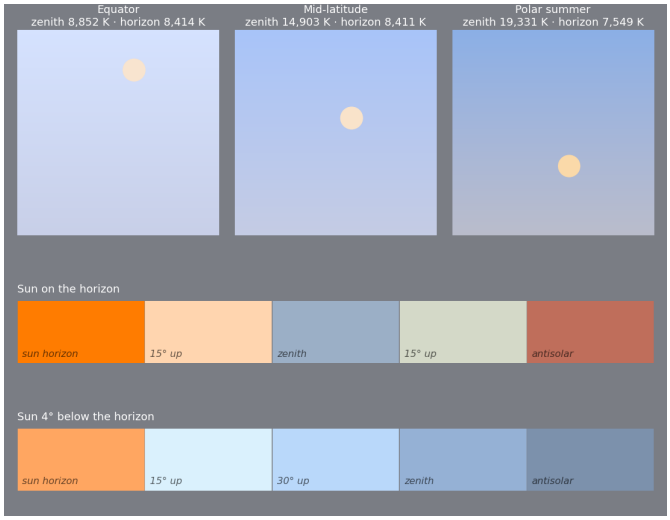


Figure 32: ζ Ophiuchi supernova, ∼1.78 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

About 1.78 million years ago a star of 16 to 18 solar masses exploded in a binary in the Scorpius-Centaurus association. The explosion unbound its companion, which is now the runaway O star ζ Ophiuchi, and left a neutron star that is now the pulsar PSR B1706-16; tracing both back puts the explosion 1.78 ± 0.21 Myr ago at 107 ± 4 pc (Neuhäuser et al. 2020). Its iron-60 may be part of the ^{60}Fe of 1.5 to 3.2 Myr ago found in deep-sea sediments and crusts (Wallner et al. 2016) and on the Moon (Fimiani et al. 2016). A typical Type II-P supernova (Richardson et al. 2014) at that distance peaks near −11.6, about twice as bright as Geminga's, or −12.3 if the binary stripped the star and made it a Type Ib. We draw the star where a straight-line trace of ζ Oph's present motion puts it, near RA 16h10m, Dec −21°. The stars are moved back along their measured space motions too; Antares has since travelled about 12°, and in the sky of the time the supernova sat about 12° west of it. That trace lands at 130 to 140 pc rather than 107, so the position is approximate. From mid-northern latitudes it stays low in the south. The air is clean Early Pleistocene air.

Geminga supernova, ∼342 ka

clean Middle Pleistocene air, under a magnitude −11 supernova in Orion

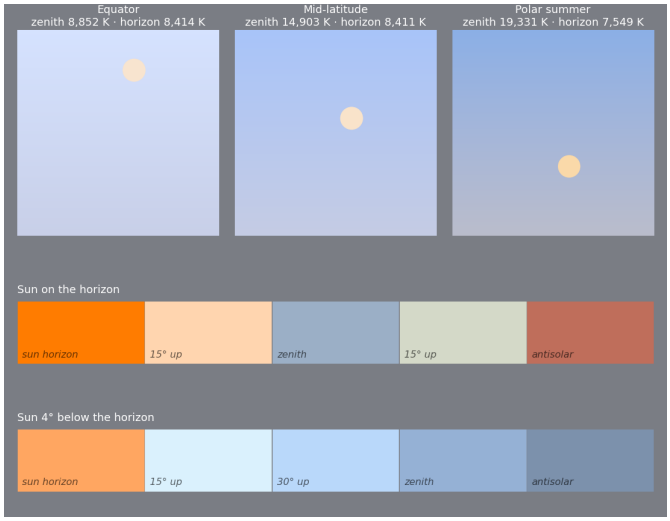


Figure 33: Geminga supernova, ∼342 ka. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

About 342,000 years ago, going by the spin-down age of the pulsar it left behind (Salvati & Sacco 2008), a star of at most 15 solar masses exploded 90 to 240 pc away, in the direction of Orion (Pellizza et al. 2005). Type II-P supernovae peak near absolute magnitude −16.75 (Richardson et al. 2014), so from the middle of that range, 147 pc, it reached about −11: a fifth of the full Moon's light, all from a point. The −13 often quoted needs both the near end of the distance range and an unusually bright explosion. It burned in the daytime sky as a dazzling star, cast sharp shadows at night, and held near peak for the three months of its plateau. The air is clean Middle Pleistocene air; at this distance the explosion barely touches the atmosphere (Thomas et al. 2016), so the sky's colors are today's. The star is drawn at the birthplace. The stars around it are moved back along their measured space motions, which puts it just 1.4° from Betelgeuse in the sky of the time.

Volcanic year (Tambora 1815 / Toba-lite)

stratospheric sulfate veil, τ_{550}∼0.4

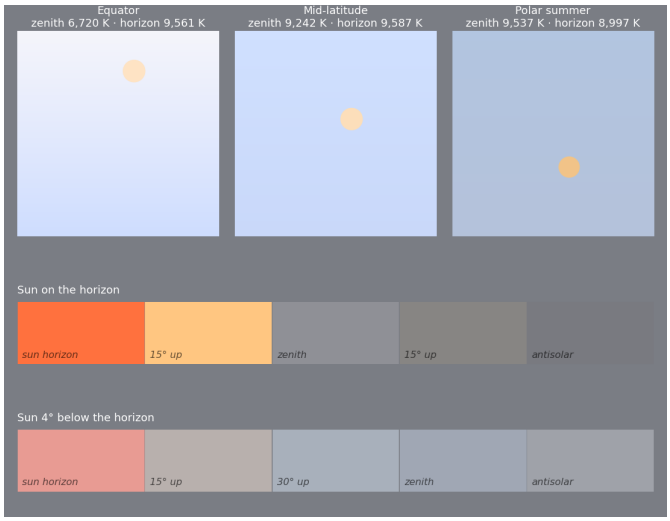


Figure 34: Volcanic year (Tambora 1815 / Toba-lite). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A Tambora-scale sulfate veil ($\tau \approx 0.4$) whitens the noon sky and lights the twilight. Noon skies are milky, whiter at the zenith than the horizon (the reverse of a clean sky). Sunsets glow salmon-orange, and the after-sunset sky 15–30° above the horizon turns a luminous pink-lavender: the high sulfate layer is still lit when the lower atmosphere is in shadow. This is the sky of Turner's 1816 paintings and of the Krakatoa twilights of 1883.

Ozone-hole spring

Antarctic spring in the ozone-hole years: column 130 DU

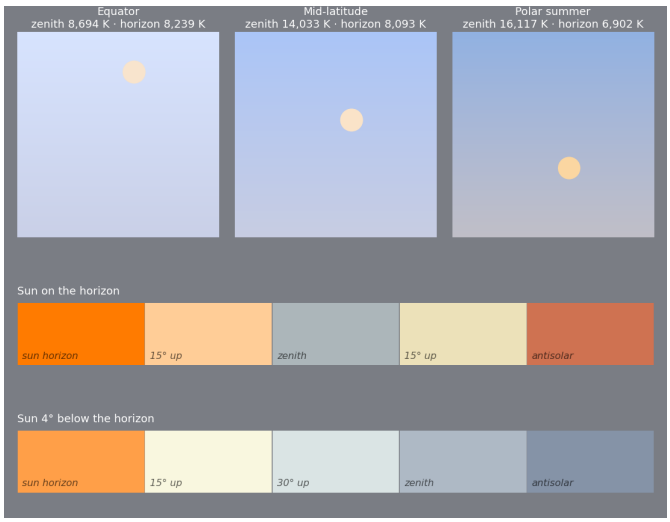


Figure 35: Ozone-hole spring. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Antarctic spring in the ozone-hole years, with the column cut to 130 DU and the layer lowered toward 18 km. Noon stays blue. Twilight is where the hole shows: with the Sun 4° down the zenith is a pale blue

(∼7,800 K) instead of the deep blue (∼12,500 K) of a 300 DU sky. It is the same direction as the thin post-oxidation column, and still well short of the cream (∼6,000 K) of a sky with no ozone at all.

Modern, clean air
aerosol optical depth 0.1; ozone 260/300/330 DU by latitude

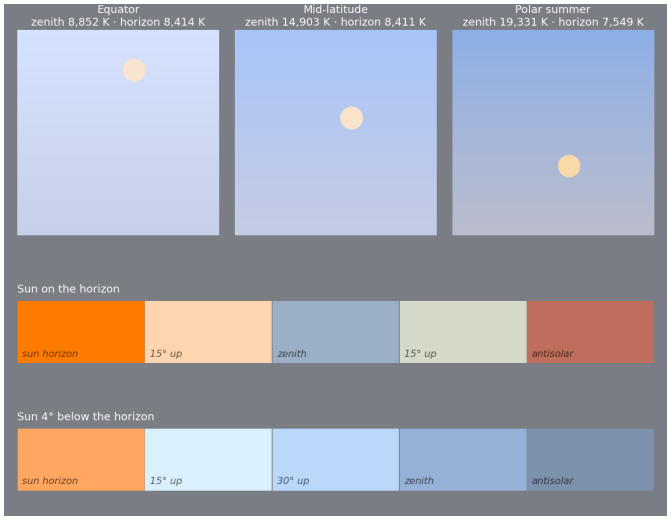


Figure 36: Modern, clean air. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Today's clean air is the baseline the other skies are measured against. The zenith is ∼9,000 K at the equator, ∼15,000 K at mid-latitudes and ∼19,000 K in polar summer, and the horizons, at 7,500–8,400 K, are essentially white. The setting Sun is ∼1,700 K, with the solar horizon orange and the antisolar horizon pink.

Modern, polluted megacity
aerosol optical depth 0.6, slightly absorbing

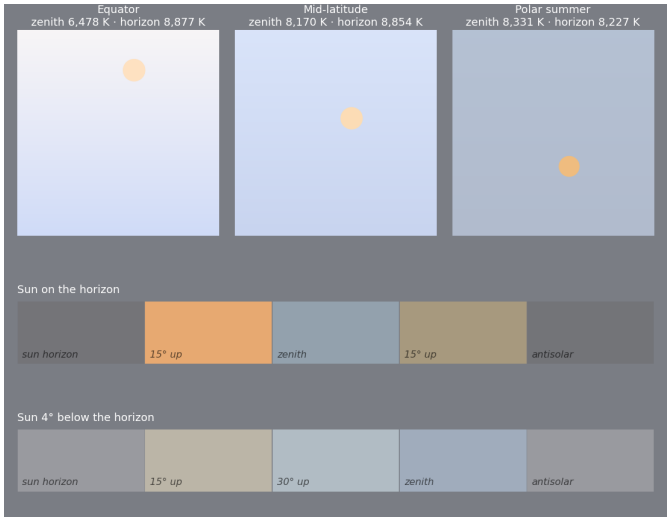


Figure 37: Modern, polluted megacity. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Aerosol optical depth 0.6 flattens the sky to a near-uniform pale gray-blue (6,500–8,300 K), brighter and whiter at the zenith than a clean sky, with the horizon slightly bluer than the zenith. The Sun disappears into gray murk well before it reaches the horizon.

Year 2100
today's clean air, with the filed megaconstellation and the Sunrise orbital datacenters

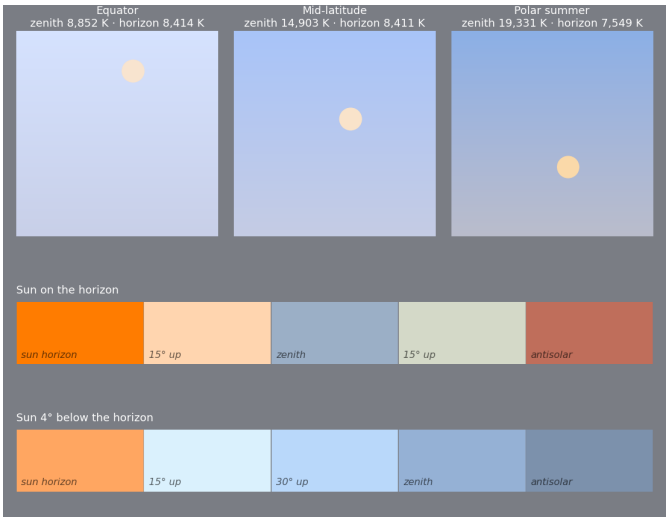


Figure 38: Year 2100. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disk in its own color. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

The air is today's. What changes is the traffic: the filed megaconstellations and, above them, the Sunrise orbital datacenters, thousands of them above naked-eye brightness an hour after sunset in a bright band along the terminator (Satellites, Section 9).

11 Caveats

Paleoatmospheric compositions carry order-of-magnitude uncertainty, and the haze optical properties are taken from Titan-analog tholins; the three haze cases bracket that uncertainty. Aerosol loads for every pre-Cenozoic epoch are educated guesses, and the proto-Earth's air before the last giant impact is unconstrained. The color-matching functions use an analytic fit, and colors are shown without chromatic adaptation, so the tints represent what a modern camera set to daylight balance would record rather than what an adapted observer would "see" (an adapted eye would perceive the hazy Archean sky as nearer to white and the modern sky as bluer than shown). Clouds are omitted from the radiative transfer throughout. The two-stream multiple-scattering term is approximate for the thickest atmospheres, where a full Monte Carlo treatment would refine the exact tint of the Hadean sky.

Beyond the colors, the strengths of the ancient airglow, aurorae, coronae, meteor, shower and comet rates and zodiacal dust are estimates set to show the trend, as each section says; the lunar eclipses, the Moon's orbit and the stars' motions back to 66 Ma are computed, and the eclipse shadow assumes today's cloud tops in every epoch. Several things are drawn for the eye rather than to scale: the Sun and Moon at four times their size (so eclipses are found and timed with the enlarged disks), the sunset's dispersion up to 24 times, and the limb darkening at a fifth of its strength. Some things cannot be known and are drawn at random: where the planets stood on a given day before history, and every star before 66 Ma, which are stand-ins with today's statistics. The Earth's axis keeps today's tilt and direction in every epoch. The Milky Way keeps today's structure, turned only to where its center lay. The scenery and the clouds are drawn, not modeled, and the invented faces of Theia and the molten Moon are illustrations of what such worlds would be like.

References

V. S. Airapetian, A. Glocer, G. Gronoff, E. Hébrard, and W. Danchi. Prebiotic chemistry and atmospheric warming of early Earth by an active young Sun. *Nature Geoscience*, 9:452–455, 2016.

E. Anderson and Ch. Francis. XHIP: An extended Hipparcos compilation. *Astronomy Letters*, 38:331–346, 2012.

Herbert Appleman. The formation of exhaust condensation trails by jet aircraft. *Bulletin of the American Meteorological Society*, 34:14–20, 1953.

Giada Arney, Shawn D. Domagal-Goldman, Victoria S. Meadows, et al. The pale orange dot: The spectrum and habitability of hazy Archean Earth. *Astrobiology*, 16:873–899, 2016.

R. Arvidson, R. Drozd, E. Guinness, C. Hohenberg, C. Morgan, R. Morrison, and V. Oberbeck. Cosmic ray exposure ages of Apollo 17 samples and the age of Tycho. In *Proceedings of the 7th Lunar Science Conference*, pages 2817–2832, 1976.

Isabelle Baraffe, Derek Homeier, France Allard, and Gilles Chabrier. New evolutionary models for pre-main sequence and main sequence low-mass stars down to the hydrogen-burning limit. *Astronomy & Astrophysics*, 577:A42, 2015.

J. R. Barnes et al. The dependence of differential rotation on temperature and rotation. *Monthly Notices of the Royal Astronomical Society*, 357:L1–L5, 2005. doi: 10.1111/j.1745-3933.2005.08587.x.

I. Baumann and S. K. Solanki. On the size distribution of sunspot groups in the Greenwich sunspot record 1874–1976. *Astronomy & Astrophysics*, 443:1061–1066, 2005.

S. Baumbach. Strahlung, Ergiebigkeit und Elektronendichte der Sonnenkorona. *Astronomische Nachrichten*, 263:121–134, 1937.

K. J. Becker, M. S. Robinson, T. L. Becker, et al. First global digital elevation model of Mercury. In *47th Lunar and Planetary Science Conference*, page 2959, 2016.

Aaron C. Boley, Samantha M. Lawler, and Hanno Rein. Rings in the sky: orbital data centres and potential impacts to astronomy and the sky. *arXiv e-prints*, page arXiv:2608.02757, 2026.

Jiří Borovička, Pavel Spurný, Peter Brown, et al. The trajectory, structure and origin of the Chelyabinsk asteroidal impactor. *Nature*, 503:235–237, 2013.

Juan M. Borrero and Kiyoshi Ichimoto. Magnetic structure of sunspots. *Living Reviews in Solar Physics*, 8:4, 2011.

Edward Bowell, Bruce Hapke, Deborah Domingue, et al. Application of photometric models to asteroids. In *Asteroids II*, pages 524–556. University of Arizona Press, 1989.

Duarte Branco, Pedro Machado, and Sean N. Raymond. Dynamical origin of Theia, the last giant impactor on Earth. *Icarus*, 441:116724, 2025. doi: 10.1016/j.icarus.2025.116724.

Marina Brozović, Robert A. Jacobson, and Ryan S. Park. Revised ephemerides of the Martian satellites, Phobos and Deimos. *Astronomical Journal*, 2025. doi: 10.3847/1538-3881/addab4.

Adam Burrows, Mark Marley, William B. Hubbard, et al. A nongray theory of extrasolar giant planets and brown dwarfs. *Astrophysical Journal*, 491:856–875, 1997.

David C. Catling and Kevin J. Zahnle. The Archean atmosphere. *Science Advances*, 6: eaax1420, 2020.

Erandi Chavez, Erin Redwing, Imke de Pater, et al. Drift rates of major Neptunian features between 2018 and 2021. *Icarus*, 404:115667, 2023.

P. Cinzano, F. Falchi, and C. D. Elvidge. The first World Atlas of the artificial night sky brightness. *Monthly Notices of the Royal Astronomical Society*, 328:689–707, 2001.

Gareth S. Collins, H. Jay Melosh, and Robert A. Marcus. Earth impact effects program: a web-based computer program for calculating the regional environmental consequences of a meteoroid impact on Earth. *Meteoritics & Planetary Science*, 40:817–840, 2005.

Gregory J. Cooke, Daniel R. Marsh, Catherine Walsh, Benjamin Black, and Jean-Francois Lamarque. A revised lower estimate of ozone columns during Earth's oxygenated history. *Royal Society Open Science*, 9:211165, 2022. doi: 10.1098/rsos.211165.

Matija Ćuk and Sarah T. Stewart. Making the Moon from a fast-spinning Earth: a giant impact followed by resonant despinning. *Science*, 338:1047–1052, 2012.

Matija Ćuk, Douglas P. Hamilton, Simon J. Lock, and Sarah T. Stewart. Tidal evolution of the Moon from a high-obliquity, high-angular-momentum Earth. *Nature*, 539:402–406, 2016.

Matija Ćuk, David A. Minton, Jennifer L. L. Pouplin, and Carlisle Wishard. Evidence for a past Martian ring from the orbital inclination of Deimos. *Astrophysical Journal Letters*, 896:L28, 2020. doi: 10.3847/2041-8213/ab974f.

Matija Ćuk, Kaustub Anand, and David A. Minton. Two possible orbital histories of Phobos. *Planetary Science Journal*, 2025. doi: 10.3847/psj/adc1ba.

Jeffrey N. Cuzzi and Paul R. Estrada. Compositional evolution of Saturn's rings due to meteoroid bombardment. *Icarus*, 132:1–35, 1998.

Jeffrey N. Cuzzi, Jack J. Lissauer, Larry W. Esposito, et al. Saturn's rings: properties and processes. In Richard Greenberg and André Brahic, editors, *Planetary Rings*, pages 73–199. University of Arizona Press, 1984.

Houraa Daher, Brian K. Arbic, James G. Williams, et al. Long-term Earth-Moon evolution with high-level orbit and ocean tide models. *Journal of Geophysical Research: Planets*, 126:e2021JE006875, 2021.

Brynna G. Downey, Francis Nimmo, and Isamu Matsuyama. The thermal-orbital evolution of the Earth-Moon system with a subsurface magma ocean and fossil figure. *Icarus*, 389:115257, 2023. doi: 10.1016/j.icarus.2022.115257.

Emil Dziewanowski. Flowfields. https://emildziewanowski.com/flowfields/, 2024. Accessed 11 October 2026.

Bengt Edlén. The refractive index of air. *Metrologia*, 2:71–80, 1966.

Graham Harper Edwards, C. Brenhin Keller, Elisabeth R. Newton, and Cameron W. Stewart. An early giant planet instability recorded in asteroidal meteorites. *Nature Astronomy*, 2024. doi: 10.1038/s41550-024-02340-6.

Fred Espenak and Jean Meeus. Five millennium canon of lunar eclipses: -1999 to +3000. Technical Report TP-2009-214172, NASA, 2009.

Larry W. Esposito, André Brahic, Joseph A. Burns, and Essam A. Marouf. Particle properties and processes in Uranus' rings. In Jay T. Bergstralh, Ellis D. Miner, and Mildred S. Matthews, editors, *Uranus*, pages 410–465. University of Arizona Press, 1991.

Tom Eulenfeld and Christoph Heubeck. Constraints on Moon's orbit 3.2 billion years ago from tidal bundle data. *Journal of Geophysical Research: Planets*, 128: e2022JE007466, 2023.

Fabio Falchi et al. The new world atlas of artificial night sky brightness. *Science Advances*, 2:e1600377, 2016.

Mohammad Farhat, Pierre Auclair-Desrotour, Gwenaël Boué, and Jacques Laskar. The resonant tidal evolution of the Earth-Moon distance. *Astronomy & Astrophysics*, 665: L1, 2022.

Y. I. Feldstein and G. V. Starkov. Dynamics of auroral belt and polar geomagnetic disturbances. *Planetary and Space Science*, 15:209–229, 1967.

Alexei V. Filippenko. Optical spectra of supernovae. *Annual Review of Astronomy and Astrophysics*, 35:309–355, 1997.

L. Fimiani et al. Interstellar ^{60}Fe on the surface of the Moon. *Physical Review Letters*, 116:151104, 2016.

Leigh N. Fletcher, Imke de Pater, Glenn S. Orton, et al. Ice giant circulation patterns: implications for atmospheric probes. *Space Science Reviews*, 216:21, 2020.

Jonathan J. Fortney, Masahiro Ikoma, Nadine Nettelmann, Tristan Guillot, and Mark S. Marley. Self-consistent model atmospheres and the cooling of the Solar System's giant planets. *Astrophysical Journal*, 729:32, 2011.

Roger Fouquet and Peter J. G. Pearson. Seven centuries of energy services: The price and use of light in the United Kingdom (1300–2000). *The Energy Journal*, 27:139–177, 2006.

Richard G. French, Matthew M. Hedman, Philip D. Nicholson, Pierre-Yves Longaretti, and Colleen A. McGhee-French. The Uranus system from occultation observations (1977–2006): rings, pole direction, gravity field, and masses of Cressida, Cordelia, and Ophelia. *Icarus*, 411:115957, 2024.

Gaia Collaboration, A. Vallenari, et al. Gaia Data Release 3: Summary of the content and survey properties. *Astronomy & Astrophysics*, 674:A1, 2023.

Florian Gallet and Jérôme Bouvier. Improved angular momentum evolution model for solar-like stars. *Astronomy & Astrophysics*, 556:A36, 2013. doi: 10.1051/0004-6361/201321302.

Enrique García-Melendo, Santiago Pérez-Hoyos, Agustín Sánchez-Lavega, and Ricardo Hueso. Saturn's zonal wind profile in 2004–2009 from Cassini ISS images and its long-term variability. *Icarus*, 215:62–74, 2011. doi: 10.1016/j.icarus.2011.07.005.

A. García Muñoz, E. Pallé, M. R. Zapatero Osorio, and E. L. Martín. The impact of the Kasatochi eruption on the Moon's illumination during the August 2008 lunar eclipse. *Geophysical Research Letters*, 38:L14805, 2011. doi: 10.1029/2011GL047981.

A. García Muñoz, M. R. Zapatero Osorio, R. Barrena, P. Montañés-Rodríguez, E. L. Martín, and E. Pallé. Glancing views of the Earth: from a lunar eclipse to an exoplanetary transit. *Astrophysical Journal*, 755:103, 2012.

Klaus Gierens, Ulrich Schumann, Manfred Helten, Herman Smit, and Alain Marenco. A distribution law for relative humidity in the upper troposphere and lower stratosphere derived from three years of MOZAIC measurements. *Annales Geophysicae*, 17:1218–1226, 1999.

Samuel Glasstone and Philip J. Dolan. *The Effects of Nuclear Weapons*. US Department of Defense and Energy Research and Development Administration, 3rd edition, 1977.

D. A. Godfrey. A hexagonal feature around Saturn's north pole. *Icarus*, 76:335–356, 1988. doi: 10.1016/0019-1035(88)90075-9.

Colin Goldblatt, Jake K. Eager-Nash, and Julia E. Horne. Evolution of the Archean atmosphere. arXiv:2409.13105, 2024. In: The Archean Earth, Elsevier (in press).

Manuel Güdel, Edward F. Guinan, and Stephen L. Skinner. The X-ray Sun in time: a study of the long-term evolution of coronae of solar-type stars. *Astrophysical Journal*, 483:947–960, 1997.

Rudolf A. Hanel, Barney J. Conrath, L. W. Herath, Virgil G. Kunde, and Joseph A. Pirraglia. Albedo, internal heat, and energy balance of Jupiter: preliminary results of the Voyager infrared investigation. *Journal of Geophysical Research*, 86:8705–8712, 1981.

Rudolf A. Hanel, Barney J. Conrath, Virgil G. Kunde, John C. Pearl, and Joseph A. Pirraglia. Albedo, internal heat flux, and energy balance of Saturn. *Icarus*, 53:262–285, 1983.

G. M. Harper, A. Brown, E. F. Guinan, E. O'Gorman, A. M. S. Richards, P. Kervella, and L. Decin. An updated 2017 astrometric solution for Betelgeuse. *Astronomical Journal*, 154:11, 2017. doi: 10.3847/1538-3881/aa6ff9.

David H. Hathaway. A standard law for the equatorward drift of the sunspot zones. *Solar Physics*, 273:221–230, 2011.

David H. Hathaway, Robert M. Wilson, and Edwin J. Reichmann. The shape of the sunspot cycle. *Solar Physics*, 151:177–190, 1994.

Andrew J. Hesselbrock and David A. Minton. An ongoing satellite–ring cycle of Mars and the origins of Phobos and Deimos. *Nature Geoscience*, 10:266–269, 2017.

D. Hestroffer and C. Magnan. Wavelength dependency of the Solar limb darkening. *Astronomy & Astrophysics*, 333:338–342, 1998.

H. Hiesinger, J. W. Head, U. Wolf, R. Jaumann, and G. Neukum. Ages and stratigraphy of lunar mare basalts: A synthesis. In *Recent Advances and Current Research Issues in Lunar Stratigraphy*, volume 477 of *GSA Special Paper*, pages 1–51. 2011. doi: 10.1130/2011.2477(01).

Dorrit Hoffleit and Jr. Warren, Wayne H. The Bright Star Catalogue, 5th revised edition. Yale University Observatory; VizieR catalogue V/50, 1991.

Andrew I. Hsu, Michael H. Wong, and Amy A. Simon. Lifetimes and occurrence rates of dark vortices on Neptune from 25 years of Hubble Space Telescope images. *Astronomical Journal*, 157:152, 2019.

Luciano Iess, Burkhard Militzer, Yohai Kaspi, et al. Measurement and implications of Saturn's gravity field and ring mass. *Science*, 364:eaat2965, 2019.

Patrick G. J. Irwin, Daniel Toledo, Ryan Garland, et al. Detection of hydrogen sulfide above the clouds in Uranus's atmosphere. *Nature Astronomy*, 2:420–427, 2018.

Patrick G. J. Irwin, Jack Dobinson, Arjuna James, et al. Modelling the seasonal cycle of Uranus's colour and magnitude, and comparison with Neptune. *Monthly Notices of the Royal Astronomical Society*, 527:11521–11538, 2024.

Luigi G. Jacchia, Franco Verniani, and Robert E. Briggs. An analysis of the atmospheric trajectories of 413 precisely reduced photographic meteors. *Smithsonian Contributions to Astrophysics*, 10:1–139, 1967.

Peter Jenniskens. Meteor stream activity. I. The annual streams. *Astronomy and Astrophysics*, 287:990–1013, 1994.

Peter Jenniskens. Meteor stream activity. II. Meteor outbursts. *Astronomy and Astrophysics*, 295:206–235, 1995.

Richard G. Jerousek, Joshua E. Colwell, Larry W. Esposito, et al. Small particles and self-gravity wakes in Saturn's rings from UVIS and VIMS stellar occultations. *Icarus*, 279:36–50, 2016.

Meridith Joyce, Shing-Chi Leung, László Molnár, Michael Ireland, Chiaki Kobayashi, and Ken'ichi Nomoto. Standing on the shoulders of giants: new mass and distance estimates for Betelgeuse through combined evolutionary, asteroseismic, and hydrodynamic simulations with MESA. *Astrophysical Journal*, 902:63, 2020. doi: 10.3847/1538-4357/abb8db.

F. D. Kahn and L. Woltjer. Intergalactic matter and the Galaxy. *Astrophysical Journal*, 130:705–717, 1959.

Nathan A. Kaib and Sean N. Raymond. A potential signature of HD 7977's passage among observed long-period comet orbits. *Planetary Science Journal*, 7:160, 2026. doi: 10.3847/PSJ/ae7a65.

Nitya Kallivayalil, Roeland P. van der Marel, Gurtina Besla, et al. Third-epoch Magellanic Cloud proper motions. I. Hubble Space Telescope/WFC3 data and orbit implications. *Astrophysical Journal*, 764:161, 2013.

Lisa Kaltenegger, Zifan Lin, and Jack Madden. High-resolution transmission spectra of Earth through geological time. *Astrophysical Journal Letters*, 892:L17, 2020.

Erich Karkoschka. Neptune's cloud and haze variations 1994–2008 from 500 HST–WFPC2 images. *Icarus*, 215:759–773, 2011.

Maria Katsova. The evolution of the solar-stellar activity. *Journal of Atmospheric and Solar-Terrestrial Physics*, 211:105456, 2020. doi: 10.1016/j.jastp.2020.105456.

Richard A. Keen. Volcanic aerosols and lunar eclipses. *Science*, 222:1011–1013, 1983.

Sascha Kempf, Nicolas Altobelli, Jürgen Schmidt, et al. Micrometeoroid infall onto Saturn's rings constrains their age to no more than a few hundred million years. *Science Advances*, 9:eadf8537, 2023.

Elise W. Knutsen et al. Detection of visible-wavelength aurora on Mars. *Science Advances*, 11:eads1563, 2025. doi: 10.1126/sciadv.ads1563.

Jun Korenaga. Rapid solidification of Earth's magma ocean limits early lunar recession. *Icarus*, 400:115564, 2023.

V. A. Krasnopolsky. Venus spectroscopy in the 3000–8000 å region by Veneras 9 and 10. In *Venus*, pages 459–483. University of Arizona Press, 1983.

Kevin Krisciunas and Bradley E. Schaefer. A model of the brightness of moonlight. *Publications of the Astronomical Society of the Pacific*, 103:1033–1039, 1991.

Christopher C. M. Kyba, Thomas Ruhtz, Jürgen Fischer, and Franz Hölker. Cloud coverage acts as an amplifier for ecological light pollution in urban ecosystems. *PLoS ONE*, 6:e17307, 2011.

Christopher C. M. Kyba et al. Citizen scientists report global rapid reductions in the visibility of stars from 2011 to 2022. *Science*, 379:265–268, 2023.

Jacques Laskar. A numerical experiment on the chaotic behaviour of the Solar System. *Nature*, 338:237–238, 1989.

Samantha M. Lawler, Aaron C. Boley, and Hanno Rein. Visibility predictions for near-future satellite megaconstellations: latitudes near 50 degrees will experience the worst light pollution. *Astronomical Journal*, 163:21, 2022.

Raymond L. Lee and Duncan C. Mollner. Tropospheric haze and colors of the clear twilight sky. *Applied Optics*, 56:G179, 2017.

Ch. Leinert et al. The 1997 reference of diffuse night sky brightness. *Astronomy and Astrophysics Supplement Series*, 127:1–99, 1998.

František Link. *Eclipse Phenomena in Astronomy*. Springer, 1969.

Beibei Liu, Sean N. Raymond, and Seth A. Jacobson. Early Solar System instability triggered by dispersal of the gaseous disk. *Nature*, 604:643–646, 2022.

David K. Lynch and Steven Mazuk. On the colors of distant objects. *Applied Optics*, 44: 5737, 2005.

M. T. Mak et al. 3D simulations of the Archean Earth including photochemical haze profiles. *Journal of Geophysical Research: Atmospheres*, 128:e2023JD039343, 2023. doi: 10.1029/2023JD039343.

Anthony Mallama and James L. Hilton. Computing apparent planetary magnitudes for The Astronomical Almanac. *Astronomy and Computing*, 25:10–24, 2018.

S. Marchi et al. Widespread mixing and burial of Earth's Hadean crust by asteroid impacts. *Nature*, 511:578–582, 2014.

Essam A. Marouf et al. Cassini radio occultation by Saturn's rings, Rev 7 egress, 3 May 2005. NASA Planetary Data System, Ring-Moon Systems Node, volume CORSS_8001, n.d.

Michaël Marsset, Miroslav Brož, Pierre Vernazza, et al. The violent collisional history of aqueously evolved (2) Pallas. *Nature Astronomy*, 4:569–576, 2020.

Matthias M. M. Meier, Andreas Reufer, and Rainer Wieler. On the origin and composition of Theia: constraints from new models of the Giant Impact. *Icarus*, 242:316–328, 2014.

Patrick Minnis, David F. Young, Donald P. Garber, et al. Transformation of contrails into cirrus during SUCCESS. *Geophysical Research Letters*, 25:1157–1160, 1998.

Brett M. Morris. A relationship between stellar age and spot coverage. *Astrophysical Journal*, 893:67, 2020. doi: 10.3847/1538-4357/ab79a0.

N.-E. Nemec et al. Faculae cancel out on the surfaces of active Suns. *Astrophysical Journal Letters*, 934:L23, 2022. doi: 10.3847/2041-8213/ac8155.

David Nesvorný. Young Solar System's fifth giant planet? *Astrophysical Journal Letters*, 742:L22, 2011.

David Nesvorný and Alessandro Morbidelli. Statistical study of the early Solar System's instability with four, five, and six giant planets. *Astronomical Journal*, 144:117, 2012.

David Nesvorný et al. Cometary origin of the zodiacal cloud and carbonaceous micrometeorites. Implications for hot debris disks. *Astrophysical Journal*, 713:816–836, 2010.

Ralph Neuhäuser, Frank Gießler, and Valeri V. Hambaryan. A nearby recent supernova that ejected the runaway star ζ Oph, the pulsar PSR B1706-16, and ^{60}Fe found on Earth. *Monthly Notices of the Royal Astronomical Society*, 498:899, 2020. doi: 10.1093/mnras/stz2629.

Gerhard Neukum, Boris A. Ivanov, and William K. Hartmann. Cratering records in the inner Solar System in relation to the lunar reference system. *Space Science Reviews*, 96:55–86, 2001.

Fiona Nichols-Fleming and Eric G. Blackman. Determination of starspot covering fraction as a function of stellar age from observational data. *Monthly Notices of the Royal Astronomical Society*, 491:2706–2717, 2020. doi: 10.1093/mnras/stz3197.

Petra Odert, Helmut Lammer, Nikolai V. Erkaev, et al. Escape and fractionation of volatiles and noble gases from Mars-sized planetary embryos and growing protoplanets. *Icarus*, 307:327–346, 2018.

James O'Donoghue, Luke Moore, Jack Connerney, et al. Observations of the chemical and thermal response of 'ring rain' on Saturn's ionosphere. *Icarus*, 322:251–260, 2019.

E. Pallé, M. R. Zapatero Osorio, R. Barrena, P. Montañés-Rodríguez, and E. L. Martín. Earth's transmission spectrum from lunar eclipse observations. *Nature*, 459:814–816, 2009.

Ryan S. Park, Alexander S. Konopliv, Bruce G. Bills, et al. A partially differentiated interior for (1) Ceres deduced from its gravity field and shape. *Nature*, 537:515–517, 2016.

John C. Pearl and Barney J. Conrath. The albedo, effective temperature, and energy balance of Neptune, as determined from Voyager data. *Journal of Geophysical Research*, 96:18921–18930, 1991.

L. J. Pellizza, R. P. Mignani, I. A. Grenier, and I. F. Mirabel. On the local birth place of Geminga. *Astronomy & Astrophysics*, 435:625, 2005. doi: 10.1051/0004-6361:20042377.

Shahram Peyvandi et al. Colorimetric analysis of outdoor illumination across varieties of atmospheric conditions. *Journal of the Optical Society of America A*, 33:1049, 2016.

Carolyn C. Porco, Robert A. West, Alfred McEwen, et al. Cassini imaging of Jupiter's atmosphere, satellites, and rings. *Science*, 299:1541–1547, 2003.

T. Reinhold, K. J. Bell, J. Kuszlewicz, S. Hekker, and A. I. Shapiro. Transition from spot to faculae domination: an alternate explanation for the dearth of intermediate Kepler rotation periods. *Astronomy & Astrophysics*, 621:A21, 2019. doi: 10.1051/0004-6361/201833754.

Jürgen Rendtel. Meteor shower calendar 2027. International Meteor Organization, IMO INFO(3-26), https://www.imo.net/files/meteor-shower/cal2027.pdf, 2027.

Dean Richardson, Robert L. Jenkins, John Wright, and Larry Maddox. Absolute-magnitude distributions of supernovae. *Astronomical Journal*, 147:118, 2014.

Raluca Rufu and Robin M. Canup. Triton's evolution with a primordial Neptunian satellite system. *Astronomical Journal*, 154:208, 2017.

Julien Salmon, Sébastien Charnoz, Aurélien Crida, and André Brahic. Long-term and large-scale viscous evolution of dense planetary rings. *Icarus*, 209:771–785, 2010.

M. Salvati and B. Sacco. The Milagro anticenter hot spots: cosmic rays from the Geminga supernova? *Astronomy & Astrophysics*, 485:527, 2008. doi: 10.1051/0004-6361:200809586.

Esther Sanromá and Enric Pallé. Reconstructing the photometric light curves of Earth as a planet along its history. *Astrophysical Journal*, 744:188, 2012. doi: 10.1088/0004-637X/744/2/188.

Ernst Schmidt. Die Entstehung von Eisnebel aus den Auspuffgasen von Flugmotoren. *Schriften der Deutschen Akademie der Luftfahrtforschung*, 44:1–15, 1941. URL https://elib.dlr.de/107948/.

Birger Schmitz et al. An extraterrestrial trigger for the mid-Ordovician ice age: Dust from the breakup of the L-chondrite parent body. *Science Advances*, 5:eaax4184, 2019.

Ralph Schönrich, James Binney, and Walter Dehnen. Local kinematics and the local standard of rest. *Monthly Notices of the Royal Astronomical Society*, 403:1829–1833, 2010.

Ulrich Schumann. Formation, properties and climatic effects of contrails. *Comptes Rendus Physique*, 6:549–565, 2005.

Anna Serdyuchenko, Victor Gorshelev, Mark Weber, Wissam Chehade, and John P. Burrows. High spectral resolution ozone absorption cross-sections – Part 2: Temperature dependence. *Atmospheric Measurement Techniques*, 7:625–636, 2014.

A. I. Shapiro et al. The variability of Sun-like stars: reproducing observed photometric trends. *Astronomy & Astrophysics*, 569:A38, 2014. doi: 10.1051/0004-6361/201323086.

W. E. Sharp, J. W. F. Lloyd, and S. M. Silverman. Zenith skylight intensity and color during the total solar eclipse of 20 July 1963. *Applied Optics*, 5:787–792, 1966.

Amy A. Simon, Fachreddin Tabataba-Vakili, Richard Cosentino, et al. Historical and contemporary trends in the size, drift, and color of Jupiter's Great Red Spot. *Astronomical Journal*, 155:151, 2018.

Bradford A. Smith, Laurence A. Soderblom, Don Banfield, et al. Voyager 2 at Neptune: imaging science results. *Science*, 246:1422–1449, 1989.

Herschel B. Snodgrass. Magnetic rotation of the solar photosphere. *Astrophysical Journal*, 270:288–299, 1983.

Sami K. Solanki. Sunspots: an overview. *Astronomy and Astrophysics Review*, 11:153–286, 2003.

David S. Spiegel and Adam Burrows. Spectral and photometric diagnostics of giant planet formation scenarios. *Astrophysical Journal*, 745:174, 2012.

Lawrence A. Sromovsky, Sanjay S. Limaye, and Patrick M. Fry. Dynamics of Neptune's major cloud features. *Icarus*, 105:110–141, 1993.

Dieter Stöffler and Graham Ryder. Stratigraphy and isotope ages of lunar geologic units: chronological standard for the inner solar system. *Space Science Reviews*, 96:9–54, 2001.

Richard B. Stothers. Stratospheric transparency derived from total lunar eclipse colors, 1801–1881. *Publications of the Astronomical Society of the Pacific*, 117:1445, 2005. doi: 10.1086/497016.

Klaus G. Strassmeier. Starspots. *Astronomy and Astrophysics Review*, 17:251–308, 2009. doi: 10.1007/s00159-009-0020-6.

Robert G. Strom et al. The inner solar system cratering record and the evolution of impactor populations. *Research in Astronomy and Astrophysics*, 15:407–434, 2015.

David Sudarsky, Adam Burrows, and Philip Pinto. Albedo and reflection spectra of extrasolar giant planets. *Astrophysical Journal*, 538:885–903, 2000.

Robert M. Suggs, Danielle E. Moser, William J. Cooke, and Ronnie J. Suggs. The flux of kilogram-sized meteoroids from lunar impact monitoring. *Icarus*, 238:23–36, 2014.

John A. Tarduno et al. Geodynamo, solar wind, and magnetopause 3.4 to 3.45 billion years ago. *Science*, 327:1238–1240, 2010.

Brian C. Thomas et al. Terrestrial effects of nearby supernovae in the early Pleistocene. *Astrophysical Journal Letters*, 826:L3, 2016.

Jocelyn Tomkin. Once and future celestial kings. *Sky & Telescope*, 95(4):59, 1998.

Andrew G. Tomkins, Erin L. Martin, and Peter A. Cawood. Evidence suggesting that Earth had a ring in the Ordovician. *Earth and Planetary Science Letters*, 646:118991, 2024.

Scott Tremaine, Jihad Touma, and Fathi Namouni. Satellite dynamics on the Laplace surface. *Astronomical Journal*, 137:3706–3717, 2009.

I. G. Usoskin, S. K. Solanki, and G. A. Kovaltsov. Grand minima and maxima of solar activity: new observational constraints. *Astronomy & Astrophysics*, 471:301–309, 2007.

Roeland P. van der Marel, Mark Fardal, Gurtina Besla, et al. The M31 velocity vector. II. Radial orbit toward the Milky Way and implied Local Group mass. *Astrophysical Journal*, 753:8, 2012.

J. M. Vaquero. Historical sunspot observations: a review. *Advances in Space Research*, 40:929–941, 2007. doi: 10.1016/j.asr.2007.01.087.

J. Hunter Waite, Rebecca S. Perryman, Mark E. Perry, et al. Chemical interactions between Saturn's atmosphere and its rings. *Science*, 362:eaat2382, 2018.

A. Wallner et al. Recent near-Earth supernovae probed by global deposition of interstellar radioactive ^{60}Fe. *Nature*, 532:69–72, 2016.

William R. Ward and Douglas P. Hamilton. Tilting Saturn. I. Analytic model. *Astronomical Journal*, 128:2501–2509, 2004.

Stephen G. Warren and Richard E. Brandt. Optical constants of ice from the ultraviolet to the microwave: A revised compilation. *Journal of Geophysical Research*, 113:D14220, 2008.

Jack Wisdom, Rola Dbouk, Burkhard Militzer, et al. Loss of a satellite could explain Saturn's obliquity and young rings. *Science*, 377:1285–1289, 2022.

Brian E. Wood et al. New mass-loss measurements from astrospheric Lyα absorption. *Astrophysical Journal Letters*, 628:L143–L146, 2005.

Chris Wyman, Peter-Pike Sloan, and Peter Shirley. Simple analytic approximations to the CIE XYZ color matching functions. *Journal of Computer Graphics Techniques*, 2(2):1–11, 2013. URL https://jcgt.org/published/0002/02/01/.

Quanzhi Ye and Peter Jenniskens. Comets and meteor showers. In *Comets III*, page 799. University of Arizona Press, 2024. doi: 10.2458/azu_uapress_9780816553631-ch024.

Kevin Zahnle, Laura Schaefer, and Bruce Fegley. Earth's earliest atmospheres. *Cold Spring Harbor Perspectives in Biology*, 2:a004895, 2010.

Appendix: computed chromaticities

Table 9: CIE 1931 chromaticity (x, y) and correlated color temperature of the noon zenith and horizon sky, and direct-Sun brightness relative to the modern equatorial Sun. The supernova epochs and the Year 2100 have today's clean air and share its rows.

Epoch	Latitude	Zenith (x, y)	CCT (K)	Horizon (x, y)	CCT (K)	Sun
Proto-Earth, ∼4.55 Ga	Equator	0.350, 0.364	4,854	0.321, 0.338	6,059	0.21
Proto-Earth, ∼4.55 Ga	Mid-latitude	0.325, 0.349	5,801	0.328, 0.346	5,683	0.16
Proto-Earth, ∼4.55 Ga	Polar summer	0.344, 0.364	5,072	0.344, 0.358	5,048	0.03
Early Hadean, ∼4.50 Ga	Equator	0.421, 0.400	3,257	0.407, 0.401	3,546	0.0084
Early Hadean, ∼4.50 Ga	Mid-latitude	0.419, 0.399	3,296	0.412, 0.400	3,433	0.0021
Early Hadean, ∼4.50 Ga	Polar summer	0.399, 0.392	3,651	0.397, 0.392	3,707	1.4e-06
Late Hadean, ∼4.0 Ga	Equator	0.303, 0.317	7,193	0.292, 0.302	8,328	0.63
Late Hadean, ∼4.0 Ga	Mid-latitude	0.275, 0.288	10,537	0.294, 0.305	8,077	0.55
Late Hadean, ∼4.0 Ga	Polar summer	0.277, 0.296	9,830	0.309, 0.320	6,795	0.25
Early Archean, ∼3.8 Ga	Equator	0.291, 0.303	8,340	0.299, 0.310	7,573	0.77
Early Archean, ∼3.8 Ga	Mid-latitude	0.264, 0.274	13,112	0.301, 0.313	7,406	0.72
Early Archean, ∼3.8 Ga	Polar summer	0.259, 0.272	14,026	0.318, 0.330	6,207	0.47
Neoarchean, ∼2.7 Ga (thin haze)	Equator	0.306, 0.323	6,936	0.301, 0.316	7,325	0.72
Neoarchean, ∼2.7 Ga (thin haze)	Mid-latitude	0.288, 0.304	8,586	0.305, 0.320	7,052	0.64
Neoarchean, ∼2.7 Ga (thin haze)	Polar summer	0.290, 0.312	8,196	0.325, 0.340	5,829	0.31
Neoarchean, ∼2.7 Ga (thick haze)	Equator	0.336, 0.355	5,348	0.324, 0.346	5,858	0.46
Neoarchean, ∼2.7 Ga (thick haze)	Mid-latitude	0.331, 0.353	5,571	0.332, 0.353	5,536	0.35
Neoarchean, ∼2.7 Ga (thick haze)	Polar summer	0.355, 0.377	4,743	0.361, 0.379	4,573	0.07
Neoarchean, ∼2.7 Ga (very thick haze)	Equator	0.386, 0.396	3,986	0.377, 0.396	4,217	0.19
Neoarchean, ∼2.7 Ga (very thick haze)	Mid-latitude	0.393, 0.405	3,884	0.391, 0.406	3,943	0.1
Neoarchean, ∼2.7 Ga (very thick haze)	Polar summer	0.429, 0.422	3,281	0.429, 0.423	3,289	0.004
Paleoproterozoic, ∼2.2 Ga	Equator	0.290, 0.303	8,424	0.295, 0.306	7,928	0.87
Paleoproterozoic, ∼2.2 Ga	Mid-latitude	0.264, 0.273	13,288	0.297, 0.308	7,780	0.81
Paleoproterozoic, ∼2.2 Ga	Polar summer	0.255, 0.267	15,479	0.311, 0.323	6,643	0.51
Snowball Earth, ∼700 Ma	Equator	0.289, 0.297	8,737	0.278, 0.288	10,229	0.99
Snowball Earth, ∼700 Ma	Mid-latitude	0.256, 0.262	16,181	0.282, 0.294	9,498	0.93
Snowball Earth, ∼700 Ma	Polar summer	0.249, 0.259	18,509	0.302, 0.319	7,234	0.62
Ordovician meteor storm, ∼466 Ma	Equator	0.290, 0.302	8,425	0.291, 0.301	8,402	0.95
Ordovician meteor storm, ∼466 Ma	Mid-latitude	0.261, 0.269	14,206	0.291, 0.301	8,400	0.87
Ordovician meteor storm, ∼466 Ma	Polar summer	0.248, 0.259	18,690	0.299, 0.310	7,593	0.5
Carboniferous, ∼300 Ma	Equator	0.290, 0.303	8,421	0.287, 0.296	8,937	0.91
Carboniferous, ∼300 Ma	Mid-latitude	0.262, 0.272	13,512	0.287, 0.296	8,930	0.82
Carboniferous, ∼300 Ma	Polar summer	0.252, 0.264	16,735	0.293, 0.302	8,178	0.41
K-Pg impact winter, 66 Ma	Equator	0.362, 0.369	4,496	0.343, 0.353	5,096	0.12
K-Pg impact winter, 66 Ma	Mid-latitude	0.360, 0.367	4,558	0.350, 0.360	4,839	0.054
K-Pg impact winter, 66 Ma	Polar summer	0.372, 0.374	4,223	0.365, 0.369	4,401	0.00038
Volcanic year (Tambora 1815 / Toba-lite)	Equator	0.310, 0.324	6,720	0.282, 0.293	9,561	0.67
Volcanic year (Tambora 1815 / Toba-lite)	Mid-latitude	0.283, 0.296	9,242	0.281, 0.293	9,587	0.53
Volcanic year (Tambora 1815 / Toba-lite)	Polar summer	0.280, 0.296	9,537	0.285, 0.299	8,997	0.13
Ozone-hole spring	Equator	0.288, 0.300	8,694	0.293, 0.303	8,239	1
Ozone-hole spring	Mid-latitude	0.261, 0.270	14,033	0.294, 0.304	8,093	0.94
Ozone-hole spring	Polar summer	0.254, 0.265	16,117	0.307, 0.319	6,902	0.57
Modern, clean air	Equator	0.287, 0.299	8,852	0.291, 0.301	8,414	1
Modern, clean air	Mid-latitude	0.259, 0.267	14,903	0.291, 0.301	8,411	0.92
Modern, clean air	Polar summer	0.247, 0.257	19,331	0.299, 0.310	7,549	0.53
Modern, polluted megacity	Equator	0.313, 0.328	6,478	0.287, 0.298	8,877	0.6
Modern, polluted megacity	Mid-latitude	0.292, 0.306	8,170	0.287, 0.299	8,854	0.45
Modern, polluted megacity	Polar summer	0.290, 0.306	8,331	0.291, 0.306	8,227	0.08